

DEDICATED PRELIM PROGRAM(DPP)**UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022****DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO****TEST BOOKLET SERIES****Test Code:****DPP**

TEST BOOKLET
GENERAL STUDIES
TEST -08

E8**Time Allowed: Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BYA CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERD: Relief of the Ocean floor Divisions of the ocean floor, Movements of ocean Water Temp. + Salinity of Ocean Water, Composition of atmosphere Structure of atmosphere, Solar Radiation, Heat Balance & Temp Atm. Circulation & Weather Systems, Planetary Winds & their Patterns Imp. climatic phenomena and its effects

Q.1) Consider the following pairs:

(List-I) Oceanic Trenches (List-II) Region

- | | | |
|-----------------------|---|-----------------|
| 1. Mariana trench | : | Atlantic ocean. |
| 2. Puerto Rico trench | : | Pacific ocean. |
| 3. Eurasia basin | : | Indian ocean. |
| 4. Java trench | : | Arctic ocean. |

Which of the pairs given above are correctly matched?

- a) 1, 2 and 4 only
- b) 1, 3 and 4 only
- c) 2, 3 and 4 only
- d) None of the above.

Q.2) The term andesite line relates to:

- a) The boundary line between continental slope and the continental shelf.
- b) The ocean ward margin of continental shelf where rapid increase in slope angle occurs.
- c) The undulating plain lying beyond the continental rise, covering two-third of the ocean floor.
- d) The area where polymetallic nodules of copper, magnesium and cobalt are found on seafloor.

Q.3) Consider the following statements with respect to oceans:

- 1. Among oceans the Atlantic Ocean covers maximum of the earth surface and Indian ocean the least surface area.
- 2. Pacific Ocean has the highest average depth followed by the Indian ocean.
- 3. Sequence of total salt contents found in oceanic water in descending order is :Sodium Chloride, Magnesium Chloride, Magnesium sulphate, Calcium sulphate, Potassium sulphate.
- 4. Oceanic surface is heavier and denser than the continental surface.

Which of the statements given above are correct?

- a) 1, 2, and 4 only
- b) 1 and 3 only
- c) 2, 3 and 4 only
- d) 1, 2, 3 and 4.

Q.4) With reference to thermohaline circulation which of the following statements is/are correct?

- 1. Driven by the differences in the density of sea water and also known as subsurface currents.
- 2. Ocean bottom relief greatly influences thermohaline circulation.
- 3. It is the mechanism behind the global conveyor belt.
- 4. Gulf stream current is an example of thermohaline circulation.

Select the Answer using the codes given below:

- a) 1, 2 and 4 only
- b) 2 and 4 only
- c) 3 only
- d) 1, 2, 3 and 4.

Q.5) Which of the following phenomenon's can be attributed to ocean currents?

- 1. Inducing heavy snowfall.
- 2. Discouraging rainfall.
- 3. Helps in breeding of marine organisms.
- 4. Hinders navigation.
- 5. Harnessing renewable source of energy.

Select the correct Answer using the codes given below:

- a) 1, 2 and 5 only
- b) 1, 2, 3 and 4 only
- c) 2, 3 and 5 only
- d) All of the above.

Q.6) Which of the following statements is/are correct regarding the temperature of the oceans?

- 1) The reduction of temperature with latitude is constant.
- 2) The temperature of the oceans varies vertically with increasing depth.

Select the Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.7) Which of the following factors are responsible for the movement of ocean currents:

- 1. The movement of planetary winds.
- 2. Temperature differences in the oceanic waters.
- 3. Variation in salinity.
- 4. The rotation of the earth.
- 5. Presence of landmass.

Select the Answer using the codes given below.

- a) 1, 2, 3 and 5 only
- b) 1, 2, 4 and 4 only
- c) 2, 3, 4 and 5 only
- d) 1, 2, 3, 4 and 5

Q.8) With reference to Indian ocean circulation, consider the following statements:

- 1. Dominant effects of winds in shaping the direction of ocean currents is most convincingly found here.
- 2. The currents of the Indian ocean form a circuit starting from and back to the equator.



Select the correct Answer from the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.9) With reference to the Coral reefs, consider the following statements:

- 1. Agricultural runoff harms the growth of coral reefs.
- 2. Coral reefs are restricted from off the coastline of South Asia from the Eastern tip of India to the Bangladesh and Myanmar borders.
- 3. Second largest barrier reef is present in Central America.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.10) Consider the following statements:

- 1. The mixing of warm and cold currents favours the growth of planktons.
- 2. Cold ocean currents have an effect on desert formation coast regions of the subtropical continents

Which of the statement(s) given above is/are incorrect?

- 1. 1 only
- 2. 2 only
- 3. Both 1 and 2
- 4. Neither 1 nor 2

Q.11) In the context of Ocean Reliefs, which of the following is/are the advantages of Continental shelves?

- 1. Provide Richest fishing Grounds
- 2. Site for fossil fuel reserves
- 3. Good source of Polymetallic
- 4. Increases the height of tides

Select the correct Answer using the code given below:

- a) 1, 2 and 3 only
- b) 2, 3 and 4 only
- c) 1, 2 and 4 only
- d) 1, 2, 3 and 4

Q.12) These are deep valleys. Some comparable to the Grand Canyon of the Colorado river

They are sometimes found cutting across the continental shelves and slopes, often extending from

the mouths of large rivers. Identify the above oceanic feature

- a) Seamount
- b) Submarine Canyons
- c) Trenches
- d) Guyots

Q.13) Consider the following statements:

- 1. Cold Currents are usually found on the east coast of continents at higher latitudes in the Northern hemisphere.
- 2. Warm Currents are usually found on the east coast of continents in the low and middle latitudes.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2 only
- d) None of the above

Q.14) Ocean Bottom floor is an outcome of

- a) Volcanic process.
- b) Volcanic and Tectonic processes only.
- c) Volcanic and Erosional processes only.
- d) Volcanic, tectonic and Erosional processes

Q.15) Consider the following statements

- 1. Continental Shelf is the shallowest part of the ocean.
- 2. The width of the continental shelves remains the same in almost all the oceans

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.16) Which statements among the following about minor ocean relief features are correct?

- a) Ridges are formed along a convergent boundary
- b) Trenches are formed along a divergent boundary
- c) Canyons are a result of depositional landform.
- d) None of the above

Q.17) Consider the following statements:

- 1. The outgoing tide along the coast and from the bays and estuaries during the time between low tide and high tide is called a flood current.
- 2. The incoming tide during the time between high tide and low tide is called an ebb current.

Which of the statements given above is/are correct?

- a) 1 only

- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.18) Which of the following factors affect the distribution of insolation:

1. Absorption.
2. Distance between the earth and the sun.
3. Reflection.
4. Angle of the sun's rays.
5. Length of the day.

Select the correct Answer from the codes given below:

- a) 1, 2, 4 and 5
- b) 2, 3 and 5
- c) 1, 2, 3 and 4
- d) 1, 2, 3, 4 and 5

Q.19) Match the following local winds with the area of occurrence:

List-I (Local Winds Names): **List-II (Region)**

- | | |
|--------------|---|
| A- Fohn | 1- Alps (Europe) |
| B- Chinook | 2- Rockies (North America) |
| C- Berg | 3- South Africa. |
| D- Khamsin | 4- Egypt. |
| E- Harmattan | 5- Guinea Coast/Eastern part of Sahara. |
| F- Samun | 6- Iran. |

Select the correct Answer using the codes given below:

- a) A-1, B-2, C-3, D-4, E-5, F-6.
- b) A-2, B-1, C-3, D-4, E-5, F-6.
- c) A-1, B-2, C-5, D-4, E-4, F-6
- d) A-2, B-1, C-5, D-4, E-6, F-3.

Q.20) Match the following cyclones with their area of occurrence.

List-I (Cyclones) **List-II (Regions)**

- | | |
|------------------|------------------------|
| A- Typhoons | 1- China sea. |
| B- Hurricanes | 2- USA. |
| C- Tornadoes | 3- Caribbean sea. |
| D- Baguio | 4- Japan. |
| E- Taifu | 5- Philippines. |
| F- Willy Willies | 6- Northern Australia. |

Select the correct Answer from the codes given below.

- a) A-1, B-2, C-3, D-4, E-5, F-6.
- b) A-1, B-3, C-2, D-5, E-4, F-6.
- c) A-2, B-1, C-2, D-5, E-4, F-6.
- d) A-2, B-1, C-3, D-4, E-5, F-6.

Q.21) Consider the following statements regarding the Jet streams:

1. The greater the horizontal temperature difference, the stronger the Jet Streams.
2. Determines local weather conditions.
3. It plays an important role in the onset and withdrawal of Indian monsoon.
4. It acts as a transporter of the radioactive dust and other pollutants.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only
- c) 2, 3 and 4
- d) 1, 2, 3 and 4

Q.22) With reference to the Life cycle of a Temperate cyclone, match the following lists:

List-I (Stages) **List-II (Characteristics)**

- | | |
|-----------------|---|
| A. First stage | 1) Warm and cold air masses penetrate into each other. |
| B. Second stage | 2) Convergence of two air masses of contrasting direction and nature. |
| C. Third stage | 3) Cyclone fully developed. |
| D. Fourth stage | 4) Warm sector is narrowed. |
| E. Fifth stage | 5) Warm sector completely disappears. |
| F. Sixth stage | 6) Occlusion of cyclone starts. |

Select the correct Answer from the codes given below:

- a) A-1, B-2, C-3, D-4, E-5, F-6.
- b) A-2, B-1, C-3, D-4, E-6, F-5.
- c) A-1, B-2, C-4, D-3, E-5, F-6.
- d) A-2, B-1, C-4, D-3, E-5, F-6.

Q.23) Consider the following statement with respect to weather system, fronts :

1. They are mostly developed in the middle latitudes and bring abrupt changes in temperature.
2. The gradient of the cold front is much steeper than that of the warm front.
3. Cold fronts move faster than the warm fronts.
4. On weather maps cold fronts are illustrated by blue lines, whereas red fronts are illustrated by red lines.

Which of the above statements is/are correct?

- a) 1, 2 and 4 only
- b) 2 only
- c) 2, 3 and 4 only

d) 1, 2, 3 and 4

Q.24) Which of the following is/are the characteristics of anticyclone?

1. Anticyclone moves clockwise in the Northern Hemisphere and anticlockwise in the Southern Hemisphere.
2. Weather becomes rainless, with clear skies.
3. Temperature of anticyclone is directly proportional to the local weather.

Select the correct Answer from the codes given below:

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.25) With respect to the retreating monsoons, consider the following statements:

- 1) It is marked by high temperature and humidity and occurs in the months of October and November.
- 2) The Punjab plains experience the withdrawal of monsoons first.

Which of the above given statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.26) Consider the following statements:

- 1) Earth's atmosphere is heated directly by the insolation of the sun.
- 2) Normal lapse rate is responsible for lesser temperature at higher altitudes.
- 3) Land portions of the Earth are subjected to greater temperature changes.

Select the correct Answer using the codes given below:

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1, 2, and 3

Q.27) Consider the following statements:

- 1) The isotherms of the Southern Hemisphere show greater deviation compared to the Northern hemisphere.
- 2) During winters in the Northern Hemisphere, the isotherms bend southwards along the continents.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.28) Consider the following conditions:

- 1) Fogs in winters.
- 2) Air drainage
- 3) Distance from sea level
- 4) Winter nights with clear sky

Which of the above conditions are ideal for temperature inversion to take place?

- a) 1, 2, and 4 only
- b) 1, 2, and 3 only
- c) 2, 3, and 4 only
- d) 1, 2, 3, and 4

Q.29) Which of the statements given below is most appropriate regarding the forces acting on the air?

- a) Pressure gradient force is weak in places of closer isobars.
- b) Frictional force does not cause any effect in the speed of the winds over landmasses.
- c) The Coriolis force deflects the wind to the right in the southern hemisphere.
- d) The magnitude of Coriolis force is directly proportional to the latitudinal angle.

Q.30) Consider the following statements:

- 1) Geostrophic winds flow in the absence of the force of friction.
- 2) Pressure gradient, Coriolis, and frictional force are the only forces acting on the wind.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.31) Consider the following events:

- 1) Rotation of Earth.
- 2) Oscillating pressure belts.
- 3) Distribution of land and sea.
- 4) Gravitational pull of the moon.
- 5) Sunspot cycle.

Which of the above factors is/are responsible for the distribution of the planetary winds?

- a) 1, 2, and 4 only
- b) 2, 4, and 5 only
- c) 1, 2, and 3 only



d) 2, 3, and 5 only

Q.32) Consider the following statements:

- 1) The Trade winds blow towards the Equator while the westerlies blow towards the poles.
- 2) Inter-Tropical Convergence zone is created at the convergence of Westerlies and the Trade winds.
- 3) The air flow around the equator is the Easterlies.

Select the correct Answer using the codes given below:

- a) 3 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.33) Consider the following pairs:

Pressure cells	Location
1) Hadley Cell	Subtropical high
2) Ferrel Cell	Tropical low
3) Polar Cell	Polar High

Which of the above pairs is/are incorrectly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.34) Consider the following statements:

- 1) The atmospheric air circulation plays an important role in the transfer of energy to oceans.
- 2) There is a unidirectional transfer of energy from the atmospheric air to the ocean currents.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.35) Consider the following statements:

- 1) The El Nino event is marked by the absence of cool Peruvian currents.
- 2) Strong Tropical Jet streams over Tibet are responsible for strong monsoons in India.
- 3) Southern Oscillations are marked by the changes around Indian Ocean north of Australia.

Select the correct Answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 only
- d) 3 only

Q.36) Consider the following statements:

- 1) During afternoons, pressure gradient exists from land to sea.
- 2) Warm dry winds along the leeward side of the mountains are referred to as Katabatic winds.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.37) Consider the following statements regarding fronts:

- 1) Warm fronts are formed when a warm air mass pushes and uplifts the colder one.
- 2) Fronts are the areas of sharp changes in temperature and pressure.
- 3) Occluded fronts are formed when two cold air masses uplift a comparatively cooler air mass.

Select the correct Answer using the codes given below:

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3 only

Q.38) Which among the following statements is most appropriate regarding the Extra tropical cyclones?

- a) They are thermally induced low pressure areas which move from west to east.
- b) They are initiated with the creation of an occluded front in the mid latitudes.
- c) They cover a small geographical area and inflict maximum damage to it.
- d) These cyclones have the capacity to originate on the land surface.

Q.39) Consider the following statements regarding tropical cyclones:

- 1) They are thermally induced low-pressure systems travelling in the eastward direction.
- 2) They can originate on the surface of a water body only.
- 3) They do not require any prior low pressure in the region of their formation.
- 4) The tropical cyclones formed at the Equator are most destructive in nature.

Select the incorrect Answer using the codes given below:

- a) 2, 3 and 4 only
- b) 1, 3, and 4 only
- c) 2 and 3 only
- d) 1, 2, 3 and 4

Q.40) Consider the following statements regarding cyclones in India:

- 1) Frequency of cyclones is more in the Bay of Bengal while destructivity is more of the Arabian sea cyclones.
- 2) The lack of influx of freshwater on the Arabian Sea is an important reason for its low cyclonic activity.
- 3) Occurrence of Simultaneous cyclones is responsible for the rise in the Accumulated Cyclone Energy.

Select the correct statements using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) None of the Above

Q.41) Consider the following statements:

- 1) The World Meteorological Organization and the UNESCAP initiated the system for naming cyclones.
- 2) Arabian sea is a closed sea with favourable conditions for cyclogenesis.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.42) Consider the following statements:

- 1) Warm air fronts are more frequent in the areas around the Equatorial forests.
- 2) The eye of a cyclone is an intense high-pressure area with subsiding air.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.43) Consider the following statements:

- 1) The anticyclonic winds in the Southern hemisphere move in the anticlockwise direction.
- 2) The pressure gradient force is greater horizontally than vertically.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only

- c) Both 1 and 2
- d) Neither 1 nor 2

Q.44) Consider the following statements:

1. Tide is a periodic rise and fall of sea level once or twice a day, mainly because of the pull of the moon or the sun.
2. Surges are movement of water caused by winds and atmospheric pressure changes.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.45) Which of the following factors can affect the temperature distribution in the oceans?

1. Latitude
2. Distribution of sea and water
3. Prevailing wind
4. Ocean current

Select the correct Answer using the codes given below:

- a) 1 and 2 only
- b) 3 and 4 only
- c) 1,3, and 4 only
- d) 1, 2,3 and 4

Q.46) With reference to the vertical and horizontal distribution of water, consider the following statements:

1. Thermocline is a layer of water in the ocean where the temperature decreases rapidly with increase in depth.
2. The highest surface temperature of the ocean is recorded at the equator as it is directly under the sun.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.47) With reference to salinity of water, which of the following statements is/are incorrect?

1. Salinity at the surface increases by the loss of water to ice or evaporation.
2. Any change in the temperature or density influences the salinity of water in an area.

Select the correct Answer using the codes given below:

- a) 1 only

- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.48) In the context of ocean salinity, consider the following factors:

1. Evaporation and precipitation
2. Freezing and thawing of ice
3. Inflow of fresh water
4. Wind
5. Ocean currents

Which of the above factors can affect the salinity distribution in the oceans?

- a) 1, 2, 3 and 5
- b) 2, 3, 4 and 5
- c) 1, 3, 4 and 5
- d) 1, 2, 3, 4, 5

Q.49) With reference to the formation of tides, Consider the following pairs:

Types of Tides	Features
1. Semi-diurnal Tide	- One high tide and one low tide per day.
2. Diurnal Tide	- Two high and two low Tides per day.
3. Spring Tide	- Occur when the sun, the moon and the earth are in a straight line.
4. Neap Tide	- Occur when the Sun and the Moon are at right angles to each other.

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4 only
- d) 1 and 4 only

Q.50) Consider the following statements about the movement of ocean water:

1. Oceanic circulation pattern roughly corresponds to earth's atmospheric circulation pattern.
2. A current is usually strongest at the surface and decreases in strength with depth.
3. Warm currents are usually observed on the West coast of continents in the low and middle latitudes.

Select the correct Answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.51) Consider the following statements about the composition of atmosphere:

- 1) It consists of gases only.
- 2) It consists of water vapour and gases only.
- 3) It consists of gases, water vapour, and particles of dust (solid and liquid).
- 4) The proportion of gases remains the same throughout all the layers of the atmosphere.

Which of the statements given above is/are correct?

- a) 1 and 4 only
- b) 2 and 4 only
- c) 3 only
- d) 3 and 4 only

Q.52) Consider the following statements about Carbon dioxide:

1. Carbon dioxide is opaque to the incoming solar radiation but transparent to the outgoing terrestrial radiation.
2. The volume of other gases along with the volume of carbon dioxide has been rising in the past few decades mainly because of the burning of fossil fuels.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.53) Consider the following statements:

Nitrogen and carbon dioxide make up nearly 99% of the atmosphere and the remaining gases constitute about 1% of the atmosphere.

Ozone is found between 10 and 50 km above the earth's surface and acts as a filter and absorbs the infrared from the sun and prevents them from reaching the surface of the earth.

Which of the statements given above is/are correct?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 and 2

Q.54) Consider the following statement about the composition of atmosphere:

1. Water vapour decreases with respect to altitude and increases from equator to poles.
2. Dust particles are concentrated in the lower layers of the atmosphere and they are not found at great heights.

3. The higher concentration of dust particles is found in subtropical and temperate regions as compared to equatorial and polar regions.

Which of the statements given above is/are correct?

- a) Only 1 and 3
- b) Only 2
- c) Only 1 and 2
- d) Only 3

Q.55) Consider the following statement with respect to structure of atmosphere:

- 1. Density is highest near the surface of the earth and increases with increasing altitude.
- 2. The column of atmosphere is divided into seven different layers depending upon the temperature condition.

Which of the statements given above is/are correct?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.56) Which of the following statements is/are correct about the Troposphere?

It is the lowermost layer of the atmosphere and temperature increases with increasing height.

All changes in climate and weather take place in the Troposphere.

Select the correct Answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.57) Which are the layers of atmosphere where temperature starts increasing with increasing height?

- 1) Thermosphere
- 2) Stratosphere
- 3) Troposphere
- 4) Mesosphere

Select the correct Answer using the code given below:

- a) 1 and 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 3 and 4 only

Q.58) Consider the following statements about Exosphere:

It is the upper most layer of the atmosphere. Light gases like helium and hydrogen float into space from here.

It is located between 80-400 km above the mesopause.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.59) With reference to revolution of earth around the sun consider the following statements:

- 1. During revolution around the sun, the earth is nearest to the sun on 4th July and it is called aphelion
- 2. On 3rd January earth is farthest from the sun and it is called perihelion.

Which of the statements given above is/are correct?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.60) With reference to heating and cooling of the atmosphere, which of the following statements are Incorrect?

- 1) Conduction takes place when two bodies of unequal temperature are in contact with one another, there is a flow of energy from the cooler to warmer body.
- 2) The process of horizontal heating of the atmosphere is called convection.
- 3) The process of vertical heating is called advection

Select the correct Answer using the options given below:

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Q.61) With reference to heat budget of the earth consider the following statements:

- 1. The amount of heat received in the form of insolation equals the amount lost by the earth through terrestrial radiation.
- 2. Due to the heat budget earth neither warms up nor cools down despite the huge transfer of heat that takes place.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.62) Consider the following statements about the Global Climate Risk Index:

1. It is released by an environmental think tank "Germanwatch".
2. India has been ranked 10th most vulnerable country as per the 2020 index.
3. Germanwatch also releases the "Brown to Green" report.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) 1 and 2 only
- d) 1 and 3 only

Q.63) With reference to Jet Stream, which of the following statements is/are correct?

1. Somali jet stream strengthens the south west monsoon in India
2. During winters, an increase in thermal contrast intensifies the jet stream.
3. Heating of Tibetan plateau in summer intensify the Tropical easterly Jet Stream

Select the Answer using the codes given below:

- a) 1 and 2 only
- b) 3 only
- c) 2 and 3 only
- d) 1,2 and 3

Q.64) Considering the following statements related to Air Mass, which statements is/are incorrect?

1. Anticyclonic areas are most ideal for regions for development of air mass.
2. A pre existing Low-level disturbance is essential for formation of air masses
3. Air Mass has uniform temperature and moisture conditions.

Select the Answer using the codes given below:

- a) 1 only
- b) 1 and 3 only
- c) 2 only
- d) 1 and 2 only

Q.65) Consider the following statements related to formation of ITCZ, which of the following is/are correct?

- a) ITCZ is the result of convergence of two contrasting air masses.
- b) ITCZ is formed over a subtropical high pressure zone.

- c) Mid latitude cyclone formation is one of the characteristics of ITCZ.
- d) ITCZ is a broad rough of low pressure where air ascends.

Q.66) With reference to El-Nino, which of the following statements is/are incorrect?

1. It is characterised by further decrease in temperature along the Peru and Ecuador coast.
2. This bring drought in west pacific and more rain to south american western coast
3. El-Nino year increase production of summer crops in India

Select the Answer using the codes given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3 only

Q.67) With reference to westerlies, which of the following statement is correct?

- a) Western margin of Britain get rainfall throughout the year.
- b) Between 30- 45 deg N/S latitude wet summer and dry winter is a result of westerlies
- c) Rainfall in the Laurentian climatic region is the result of westerlies.
- d) Westerlies are responsible for the rainfall in the North western part of India throughout the year.

Q.68) With reference to air mass and Fronts, which of the following statements is/are correct?

1. Deformatory circulation is most favourable for frontogenesis
2. Convergence of two different air mass is prerequisite condition for the formation of mid latitude cyclone
3. More the gentle slope of the front more the downpour and bad weather.
4. Contrast in Temperature of two air mass is more in summer session

Select the Answer using the codes given below:

- a) 1 and 4 only
- b) 2 only
- c) 1 and 2 only
- d) 3 and 4 only

Q.69) Temperate cyclone do not form on equator because

- a) At the equator two air mass do not converge



- b) Uniform temperature condition of two air masses which converge at equator
- c) Coriolis force is maximum at the equator
- d) Doldrum at the equator inhibits the formation of cyclones

Q.70) Which among the following is/are the direct implication of shifting of wind belts?

- 1. Monsoon climate in south East Asia region, Indian sub continents and parts of Africa.
- 2. Rainfall in western part of continents within zone 30 deg to 40 deg latitude in winter season.
- 3. In the summer solstice weakened polar easterlies in northern hemisphere

Select the Answer using the codes given below: -

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.71) With reference to recent refugee crisis along India's border with Myanmar, consider the following statements:

- 1. India is not a signatory to either United Nations Refugee Convention of 1951 or its 1967 protocol
- 2. Of states sharing border with Myanmar, Arunachal Pradesh shares the longest border
- 3. No Free Movement Regime exist between India and Myanmar

Which of the above statements is/are correct?

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q.72) India's first interstate tiger translocation project was initiated at which of the following sites

- a) Kanha tiger reserves
- b) Bandhavgarh tiger reserves
- c) Satkosia tiger reserves
- d) Simlipal tiger reserves
- e)

Q.73) consider the following pairs:

Ethnic group	Native
1. Karen	Myanmar
2. Yezidis	Iraq
3. Hazaras	Afghanistan
4. Uyghurs	Pakistan

Which of the above pairs is/are correctly matched?

- a) 1, 2 and 4 only

- b) 1 and 4 only
- c) 2, 3 and 4 only
- d) 1,2 and 3 only

Q.74) Consider the following statements related to Suez Canal:

- 1. Suez Canal is an artificial waterway that opened in 1869.
- 2. Suez Canal connects the Mediterranean Sea and Gulf of Persia
- 3. It is carrying over 25% of world trade by volume.
- 4. The canal separates the African continent from Asia.

Which of the above statement/s is/are incorrect?

- a) 1 and 4 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 2 and 4 only

Q.75) The Ministry of Science and Technology recently released new guidelines for the Geo-spatial sector in India. Consider the following guidelines

- 1. Prior approvals for surveying, mapping and building applications based on Geo-spatial data have been done away.
- 2. For Indian Geo-spatial entities, there will be complete deregulation with no prior approvals or security clearances.

Which of the above statements is/are incorrect?

- a) Only 1
- b) Only 2
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.76) Consider the following statements with reference to State Election Commissioner (SEC)

- 1. Article 243K referred to the superintendence and the conduct of elections vested in the State Election Commissioner.
- 2. State Election Commissioner is to be appointed by the President
- 3. State Election Commissioner shall not be removed from his/her office except in the manner and on the like grounds as a Judge of a High Court
- 4. The State Election Commissioner oversees the election of state legislation.

Which of the above statement/s is/are correct?

- a) 1 and 2 only

- b) 2 and 3 only
- c) 3 and 4 only
- d) 1 and 3 only

Q.77) Consider the following statements about Millets:

1. The Jowar, bajra, ragi are major millet crops grown in India.
2. It contains high iron content and can fight high prevalence of anemia in Indian women of reproductive age.
3. They are the non-drought resistant crops that require high water and have a long growing season.
4. They are considered as rich in antioxidants.

Which of the statements given above is/are correct?

- a) 1, 2 and 3
- b) 2, 3 and 4
- c) 1, 2 and 4
- d) All of the above

Q.78) Consider the following statement with reference to Capital Gains Tax.

- a) Capital Gains Tax is a tax applied on growth of value of investments.
- b) The tax applies to both sold and unsold investment.
- c) Capital gains treatment only applies to "capital assets".
- d) When an asset is held for a period of 36 months or less, it is termed as a short-term capital asset and above 36 months, is considered as long-term capital asset.

Which of the above statement about Capital Gains tax is incorrect?

Q.79) Consider the following statements with reference to Termination of Pregnancy (Amendment) Bill, 2020

1. Bill increases the time period from earlier 20 weeks to 24 weeks within which abortion may be carried out.
2. The Bill allows unmarried women to terminate a pregnancy for the reason of failure of contraceptive method or device.
3. After the implementation, the bill will drastically reduce maternal mortality and morbidity rate in India.

Which of the above statement/s is/are correct?

- a) 1 and 2 only

- b) 1 and 3 only
- c) 2 and 3 only
- d) All of the above

Q.80) Consider the following pairs

India's joint Exercise name involved	Partner country
--------------------------------------	-----------------

- | | |
|---------------------|------------|
| 1. DUSTLIK II | Uzbekistan |
| 2. VAJRA PRAHAR | USA |
| 3. Nomadic elephant | Kazakhstan |
| 4. Sampriti | Bangladesh |

Find which of the above pairs are correctly matched?

- a) 1, 2 and 3
- b) 1, 2 and 4
- c) 2, 3 and 4
- d) All of the above

Q.81) P172+18, sometimes seen in news is related to which of the following?

- a) A newly discovered Quasar
- b) A newly discovered exoplanet
- c) A new apache helicopter version developed by US
- d) A missile system of Russia

Q.82) With reference to Appropriation Bill, consider the the following statements:

1. In a Westminster parliamentary system, the defeat of an appropriation bill in a parliamentary vote generally necessitates the resignation of a government.
2. Constitutionally, the government can withdraw money from the Consolidated Fund only after receiving approval from Parliament.
3. The Rajya Sabha has the power to recommend any amendments in the Appropriation Bill.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.83) With reference to various Mudras related to Buddha, consider the following pairs:

- | Mudra | Significance |
|-------------------------|--|
| 1. Hridayanjali Mudra - | It represents fearlessness |
| 2. Karana Mudra - | Warding off of evil |
| 3. Varada Mudra - | It represents the offering, welcome, charity, giving, compassion and sincerity |
| 4. Vajra Mudra - | It denotes the fiery thunderbolt-air, water, fire, earth, and metal |

Which of the above pairs is/are correctly matched?

- a) 1 and 2 Only
- b) 2, 3 and 4 Only
- c) 1, 2 and 4 Only
- d) 1, 2, 3 and 4

Q.84) With reference to Pradhan Mantri Swastha Suraksha Nidhi (PMSSN) , Consider the following statements :

- 1. It is a non-lapsable fund for Health in Public accounts credited from the proceeds of Health and Education cess.
- 2. The expenditure on schemes of the Ministry of Health and Family welfare would be incurred from PMSSN for the entire period of such scheme.

Which of the above statements is/are Correct?

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2

Q.85) With reference to Millets, Consider the following statements:

- 1. Recently, FAO (Food and Agricultural Organisation) has declared year 2023 as international year of Millets
- 2. Millets can be used both as cereal for human being and fodder for animals
- 3. Millets are photo-insensitive and have low carbon and water footprint
- 4. India is the largest producer of Millets in the world

Which of the above statements is/are correct?

- a) 1, 2 & 3 only
- b) 2, 3 & 4 only
- c) 1, 2 & 4 only
- d) 1, 2, 3 & 4

Q.86) With reference to U.S. India Artificial Intelligence (USIAI) Initiative, consider the following statements:

- 1. It is an initiative by Indo-U.S.Science and Technology Forum (IUSSTF) which will serve as a platform to discuss opportunities, challenges, and barriers for bilateral Artificial Intelligence (AI) R&D collaboration.
- 2. INDIAai, which is the National AI Portal of India, is a joint initiative of Department of Science and Technology, National e-governance division and NASSCOM (National Association of Software and Services Companies)

- 3. Ministry of Electronics and Information technology is the nodal department for IUSSTF

Which of the above statements is/are Incorrect?

- a) 1 & 2 only
- b) 2 & 3 only
- c) 3 only
- d) None of the above

Q.87) With reference to Pradhan Mantri Matru Vandana Youana (PMMVY), consider the following statements:

- 1. Objective of this scheme is to compensate partially for the wage loss of women as well as to improve health seeking behavior amongst pregnant and lactating mothers, in terms of cash incentive
- 2. Under this scheme, A beneficiary is eligible to receive benefits only once
- 3. For a women eligible for Janani Suraksha Yojana too, she would get A total sum of Rs 5000 , to be provided in three installments

Which of the above statements is/are correct?

- a) 1 & 2 only
- b) 3 only
- c) 2 & 3 only
- d) 1, 2 & 3

Q.88) With reference to Earth Hour 2021, Consider the following statements:

- 1. The theme of 2021 was 'Climate change to save Earth'
- 2. It is organised by UNEP (United Nations Environment Program)
- 3. This event includes turning off non-essential lights only

Select the correct Statement using the codes given below:

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 & 3 only
- d) None of the above

Q.89) With reference to Aluminium-Air Battery, one of the Metal-air Batteries, Consider the following statements:

- 1. It utilises oxygen in the air to produce electricity.
- 2. It is more energy dense and can be recharged more number of times compared to a Lithium-ion battery.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only

- c) Both 1 & 2
- d) Neither 1 nor 2

Q.90) With reference to the wrongful prosecution in India consider the following statements:

1. According to the article 21 of Indian constitution: No person shall be deprived of his life or personal liberty except according to due process of law.
2. Code of Criminal Procedure, 1973 (CrPC), give compensation in cases of miscarriage of justice resulting in wrongful prosecution of persons.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the Above

Q.91) According to the government's recent notification, Overseas Citizens of India (OCI) do not need to carry their old and expired passports for travel to india. With reference to OCI cards consider the following statements:

1. OCI card is a life-long visa for entry into and stay in India.
2. An OCI card is not required to be re-issued each time a new passport is issued up to 50 years of age in view of biological changes in the face of the applicant.
3. OCI card is issued to people of Indian origin globally which gives them almost all the privileges of an Indian national including sale and purchase of agricultural land.

Which of the above statements is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) All of the above

Q.92) Recently the Government unveiled The Cyber Crime Volunteer Framework as a part of cyber hygiene promotion. With the context of it consider the following statements:

1. The Ministry of Home Affairs' planned to rope in two citizens from every district in India, to flag unlawful content online.
2. There is fear that it will lead to a culture of surveillance and constant suspicion in society creating potential distrust.
3. Indian Cyber Crime Coordination Centre (I4C) scheme which will act as nodal point in the fight

against cybercrime will establish The National Cybercrime Ecosystem Management Unit

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above

Q.93) He/she was prime minister of India ; served as a Deputy Collector for twelve years; imprisoned thrice during the freedom struggle. He/she was

- a) Gulzarilal Nanda
- b) Lal Bahadur Shastri
- c) Indira Gandhi
- d) Morarji Desai

Q.94) Recently the Prime Minister of India flagged off a commemorative 'Dandi March' to launch the celebrations of the 75th year of Independence. With reference to Dandi March consider the following statements:

1. Indian protests against the salt tax began in the 19th century and remained a major contentious issue throughout the period of British rule.
2. There were arrests made that day when Gandhi and his followers picked up handfuls of salt along the shore, technically "producing" salt and breaking the law.
3. Dharasana salt works, satyagraha was led by Abbas Tyabji, a seventy-six-year-old retired judge.

Select the correct Answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 Only
- d) 1, 2 and 3

Q.95) Consider the following statements:

1. It is fed by Beas river and its tributaries.
2. This wildlife sanctuary is declared as a National wetland.
3. It is situated in the district of Kangra, Himachal Pradesh.

Select the most appropriate wildlife sanctuary on the basis of above given statements.

- a) Shikari Devi Wildlife Sanctuary.
- b) Manali Wildlife Sanctuary
- c) Pong Dam Lake Wildlife Sanctuary
- d) Chandra Taal Wildlife Sanctuary

Q.96) Recently NASA concluded that Earth is safe from asteroid for next 100 years, with reference to asteroids consider the following statements:

1. Bode's law of planetary distances predicts the missing planet between Mars and Jupiter for asteroids.
2. They are also called as minor planet
3. All asteroids orbit the Sun in elliptical orbits.

Select the correct statements using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 Only
- d) 1, 2 and 3

Q.97) Consider the following statement about heat waves:

1. The increasing frequency and intensity of heat waves observed since the later period of the 20th century has been associated with Earth's internal heat change.
2. It is a weather phenomenon characterized by high humidity and high temperature.

Which of the above statements is/are correct

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) None of the Above

Q.98) Consider the following statements with reference to Supreme court's judgement(2018) on Aadhaar

1. Aadhaar has strikes a fair balance between the right of privacy of the individual along with the right to life of the same individual as a beneficiary of government benefits
2. The use of Aadhaar by the government is justified on the doctrine of proportionality.
3. It is mandatory to link Aadhaar to file tax returns, access welfare schemes, open bank accounts, and for registration at mobile operators.
4. An Aadhaar card is not proof of citizenship, but citizens may be compulsorily required to provide it to receive welfare payments and social services.

Which of the following statement/s is/are correct:

- a) Only 1, 2 and 3
- b) Only 2, 3 and 4
- c) Only 1, 2 and 4
- d) All of the above

Q.99) Recently Brute Force Acquisition was in news is related to:

- a) Encryption of mobile data
- b) Mobile device forensic cloning
- c) private land acquisition
- d) None of these

Q.100) Consider the following statements related to Himalayan serow.

1. Himalayan serow is known to be found in the eastern, western, central Himalayas but not in the trans-Himalayan region
2. Previously assessed as vulnerable, the Himalayan serow has now been categorized as near threatened in IUCN red list.
3. It is listed under schedule III of the wildlife protection act 1972.

Which of the above statement/s is/are correct?

- a) 1 and 3
- b) 1 only
- c) 1 and 2
- d) all of the above

Q.1) Ans: D

Exp:

Trenches are the deepest parts of the ocean and are relatively steep sided, narrow basins. They occur at the bases of Continental slopes and along island arcs and are associated with active volcanoes and strong earthquakes. The deepest part in the world lies in Pacific ocean in **Mariana trench** called challenger deep, is about 11, 033 km. Whereas **Puerto Rico trench** is the deepest point of the Atlantic ocean (8.65 km), **Java trench** (7.72 km) is in Indian ocean and deepest part of arctic ocean is **Eurasia basin** (5.45 km). About 32 trenches are in Pacific, 19 in the Atlantic and 6 in Indian ocean.

Q.2) Ans: a

Exp:

- The depth of water over the continental slopes varies from 200 m to 2000 m are indented by numerous submarine canyons and mounds. The boundary line between the continental slope and the shelf is called **andesite line**. The andesite line parallels the subduction zones and deep oceanic trenches around the pacific basin. It is the most significant regional geologic distribution in the Pacific ocean basin.
- The ocean ward margin of continental shelf is called **shelf break**, where rapid increase in slope angle occurs.
- The undulating plain lying beyond the continental rise, covering two-third of the ocean floor is known as the **abyssal plain**.
- No such term is designated for the area where polymetallic nodules of copper, magnesium and cobalt are found on seafloor.

Q.3) Ans: C

Exp:

- **Statement 1 is incorrect:** Nearly 71% of earth's surface is covered by seas, among which Pacific Ocean comprises 35.4%, Atlantic Ocean comprises 18.4%, Indian Ocean 14.5% and Arctic Ocean 3%.
- **Statement 2 is correct:** The average depth of the oceans is 2.65 miles or 4.3km. Pacific ocean has the highest average depth, followed by Indian ocean, Atlantic ocean and arctic ocean.
- **Statement 3 is correct:** The salinity is the total of all the salt that are dissolved in the water. The majority of the salt is the same as table salt (Sodium Chloride-77.8%), but there are other salts as well (magnesium chloride-

10.9%, magnesium sulphate-4.7%, calcium sulphate-3.6%, potassium sulphate-2.5%). The major dissolved ions in seawater are those that exist at concentrations of at least 1mg/kg. These are from more to less abundant by mass : chloride-sodium-magnesium-sulphate-calcium-potassium.

- **Statement 4 is correct:** Continental crust is broadly granitic in composition and, with a density of about 2.7 gms per cubic centimeter, is somewhat lighter than the oceanic crust, which is basaltic in composition and has a density of about 2.9-3.0 gms per cubic cm. It is less dense than the oceanic crust (basalt richer in iron and magnesium than granite), though it is considerably thicker than the oceanic crust.

Q.4) Ans: D

Exp:

Statement 1 is correct: Winds drive ocean currents in the upper 100 metres ocean surface, but there are significant ocean currents that flow around 400 metres beneath the surface and are called subsurface currents. They are driven by differences in the density of seawater, which is controlled by temperature (thermo) and salinity (haline). The above two factors deviate in the ocean because of variation in temperature and salinity. Subsurface currents generally travel at a much slower speed (1cm/sec) when compared to surface flows.

Statement 2 is correct: The ocean bottom relief plays an important role in it due to the reason that at deeper levels, ocean current is caused by difference in density between colder, saltier water and warmer, fresher water. Because the density of water increases as it becomes colder and saltier, it sinks at high latitudes and is replaced by warm water flowing northward from the tropics. Cold water typically flows below the warmer water, but when winds blowing along coastlines deflect warm surface currents away from shore due to coriolis force, they allow cold, nutrient rich water to rise to the surface.

Statement 3 is correct: The global conveyor belt is a constantly moving system of deep ocean circulation driven by temperature and salinity. This motion is caused by a combination of thermohaline currents in the deep ocean and wind driven currents on the surface. The ocean conveyor gets its "start" in the Norwegian Sea, where warm water from the Gulf Stream heats the atmosphere in the cold northern latitudes. This loss of heat to the atmosphere makes

the water cooler and denser, causing it to sink to the bottom of the ocean. As more warm water is transported north, the cooler water sinks and moves south to make room for the incoming warm water. This cold bottom water flows south of the equator all the way down to Antarctica. Eventually, the cold bottom waters return to the surface through mixing and wind-driven upwelling, continuing the conveyor belt that encircles the globe.

Statement 4 is correct: The Gulf stream current originates in the Caribbean due to the heating of the sun, it is then proceeding north along the east coast of North America until it reaches sub-polar waters in north Atlantic. Between Greenland and Norway, the cold arctic winds cool the salt laden water almost to the freezing point. Huge amounts of the now-cold, and heavier salt water sink at that point thus following the thermohaline circulation mechanism.

Q.5) Ans: D

Exp:

Statement 1 is correct: Ocean currents circulate water worldwide, they have a significant impact on the movement of heat, energy and moisture between the oceans and atmosphere and as a result they are an important factor governing the world's weather. The **Labrador, Kurile and Falkland** cold currents are responsible for heavy snowfall in the affected areas during winters.

Statement 2 is correct: The North Atlantic Drift and Kuroshio warm currents bring in rainfall along the western coasts of Europe and eastern coasts of Japan respectively. On the other hand cold currents discourage rainfall. The Kalahari desert along the western coast of South Africa and Atacama desert along South American coast owe their existence to cold **Benguela** and **Peru** currents respectively.

Statement 3 is correct: Ocean currents play an important role in the distribution, breeding and growth of the world's sea life as it transports nutrients, planktons from one area to another which favour the growth of marine organisms. By moving heat from the equator towards the poles, ocean currents play an important role in controlling the climate which is an important factor in breeding. Many ocean species follow ocean currents to and from their feeding and

breeding grounds, for ex- Sharks, Whales and Sea turtles.

Statement 4 is correct: Ocean currents hinder navigation, the convergences of cold and warm ocean currents cause dense fogs which pose obstacles in navigation. For example, due to the convergence of warm **Kuroshio** and cold **Oyashio** currents near Japan and near Newfoundland due to the convergence of warm **Gulf stream** and cold **Labrador** current. Ocean current also impacts the speed of ships as the knowledge of ocean currents is essential for modifying the speed depending upon the path of ocean currents in order to optimize time and fuel consumption.

Statement 5 is correct: Ocean currents are also gaining significance as a possible form of alternative energy. Alternative energy sources include solar, wind, waves, tides and geothermal energy. There are even proposals to utilize ocean currents such as the Gulf Stream to utilize the temperature differential between warm surface waters and colder waters deeper in the ocean to produce energy. Because the water is dense, it carries an enormous amount of energy that could possibly be captured and converted into usable form through the use of water turbines.

Q.6) Ans: B

Exp:

Statement 1 is not correct: Generally, the mean annual temperature of the surface ocean water decreases about 70 degree F, in equatorial areas to 55 degree F, at latitudes 45 degree N and S, and drops almost to freezing points at the poles. The reduction of temperature however is not constant, because of the interference by warm and cold currents, winds and air masses. Unlike the solid earth, ocean water is mobile and the variation in the temperature between different parts of the ocean can be expected. This is the reason why the Norwegian coast, even at latitudes 60 to 70 degree N, is ice free throughout the year.

Statement 2 is correct: The temperature of the oceans varies vertically with increasing depth. It decreases rapidly for the first 1200 feet (200 fathoms), at the rate of 1 degree F for every 10 fathoms, and then more slowly until a depth of 500 fathoms is reached. Beyond this, the drop is scarcely noticeable, less than 1 degree F, for every 100 fathoms. In the ocean depths below 12000 feet, the water is uniformly cold, just a little

above freezing point. It is estimated that over 80% of all the ocean waters have a temperature between 35 degree to 40 degree F.

Q.7) Ans: D

Exp:

Statement 1 is correct: The planetary winds are permanent winds (Trade winds, Westerlies and Polar easterlies) that blow from one pressure belt to the other. Between the equator and the tropics blow the **Trade Winds** which move equatorial waters polewards and westwards and warm the eastern coast of the continents. For example, the south east Trade winds drive the south equatorial current which warms the eastern coast of Brazil as the warm Brazilian current. In temperate latitudes blow the **Westerlies**, and result in the north easterly flow of water in the northern hemisphere, so that the warm Gulf stream is driven to the western coast of Europe as the North Atlantic drift. The strongest evidence of prevailing winds on current flows is seen in the North Indian ocean. Here the direction of currents changes completely with the direction of the monsoon winds which come from the north east in winter and south west in summer.

Statement 2 is correct: The differential heating of the sun at the equator and the poles causes a difference in the temperature of ocean water. As warm water is lighter and rises, and cold water is dense and sinks, warm equatorial waters move slowly along the surface polewards, while the heavier cold water of the polar regions creep slowly along the bottom of the sea equatorwards.

Statement 3 is correct: Salinity of water varies from place to place. Waters of high salinity are denser than waters of low salinity. Hence waters of low salinity flow on the surface of waters of high salinity while waters of high salinity flow at the bottom towards waters of low salinity. For example in the Mediterranean region, there is great difference in salinity between the waters of the open Atlantic and those of the partially enclosed Mediterranean sea. The less saline water of the Atlantic flows on the surface into the Mediterranean, and this is compensated for by an outflow of denser bottom water from the Mediterranean.

Statement 4 is correct: The earth's rotation deflects freely moving objects, including ocean currents, to the right (Coriolis Force). In the Northern hemisphere this is

a clockwise direction (example: the circulation of the Gulf Stream Drift and the Canaries Current). In the southern hemisphere it is an anti-clockwise direction (example: the Brazilian current and the west Wind drift).

Statement 5 is correct: Presence of landmass obstructs and diverts a current. For instance the tip of southern Chile diverts parts of the West Wind Drift northwards as the Peruvian current.

Q.8) Ans: c

Exp:

Statement 1 is correct: In the North Indian Ocean, there is a complete reversal of the direction of currents between summer and winter, due to the changes of monsoon winds. In summer from June to October, when the dominant wind is the South-West monsoon, the currents are blown from a south-westerly direction as the south west monsoon drifts. This is reversed in winter, beginning from December, when the north east monsoon blows the currents from the north east as the north east monsoon drift. The currents of the Indian ocean, demonstrate most convincingly the dominant effects of winds on the circulation of ocean currents.

Statement 2 is correct: The currents of the South Indian ocean form a circuit. The Equatorial current, turning southwards past Madagascar as the Agulhas or the Mozambique current merges with the West Wind drift, flowing eastwards and turns equatorwards as the West Australian current.

Q.9) Ans: d

Exp:

Statement 1 is correct: Due to overuse of fertilizers for maximum production, high nutrient levels are found in runoff from agricultural areas which harms the coral reefs by encouraging the growth of various algal species. It harms the sustainable growth of the entire coral reef ecosystem by favouring the growth of one component.

Statement 2 is correct: Coral reefs are rare along the American and African west coasts. This is due primarily to upwelling and strong cold coastal currents that reduce water temperatures in these areas (respectively the Peru, Benguela and Canary streams). They are also restricted from off the coastline of South

Asia from Pakistan to Bangladesh and Myanmar due to the release of vast freshwater from various river systems.

Statement 3 is correct: The Belize Barrier Reef - second largest in the world, located in Belize, Central America. The Belize Barrier Reef is a series of coral reefs straddling the coast of Belize. It is part of Mesoamerican Barrier Reef System, which is continuous from Cancun on the north eastern tip of the Yucatan peninsula Riviera Maya and down to Honduras, making it the second largest coral reef system in the world after the Great Barrier Reef in Australia.

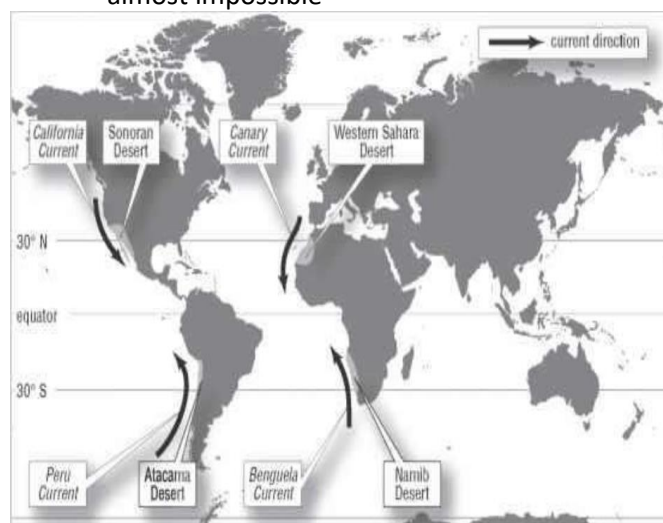
Q.10) Ans: D

Exp:

Effects of Ocean Currents

- **Statement 1 is correct:** Mixing cold and warm ocean currents bear the richest fishing grounds in the world
- The mixing of warm and cold currents helps in replenishing the oxygen and favors the growth of planktons, the primary food for fish populations.
- The best fishing grounds of the world exist mainly in these mixing zones
- **Statement 2 is correct:** Cold ocean currents have a direct effect on desert formation in west coast regions of the tropical and subtropical continents.
- They include the biggest Sahara Desert. The next biggest desert is the Great Australian Desert. The other hot deserts are the Arabian Desert, Iranian Desert, Thar Desert, Kalahari and Namib Deserts
- The hot deserts lie along the Horse Latitudes or the SubTropical High Pressure Belts where the air is descending, a condition least favorable for precipitation of any kind to take place.
- The rain-bearing Trade Winds blow off-shore and the Westerlies that are on-shore blow outside the desert limits.
- Whatever winds reach the deserts blow from cooler to warmer regions, and their relative

humidity is lowered, making condensation almost impossible



Q.11) Ans: D

Exp:

Importance of Continental Shelves

- Marine food comes almost entirely from continental shelves as they provide the richest fishing grounds.
- They are potential sites for fossil fuel reserves like petroleum and gas hydrates.
- Polymetallic nodules (manganese nodules; concentric layers of iron and manganese hydroxides), which are found over some of the continental shelves, are good sources of various mineral ores like manganese, iron copper, gold etc. Their limited depth and gentle slope keep out cold undercurrents and thereby increase the height of tides. This sometimes hinders shipping and other marine activities since ships can only enter and leave port on the tide typically ends at a very steep slope, called the shelf break.
- The continental shelves are covered with variable thicknesses of sediments brought down by rivers, glaciers etc.
- Massive sedimentary deposits received over a long time by the continental shelves, become the source of fossil fuels [Petroleum].
- The shelf is formed mainly due to submergence of a part of a continent relative rise in sea level

Q.12) Ans: B

Exp:



Submarine Canyons

These are deep valleys, some comparable to the Grand Canyon of the Colorado river. They are sometimes found cutting across the continental shelves and slopes, often extending from the mouths of large rivers. The Hudson Canyon is the best known submarine canyon in the world

Seamount

It is a mountain with pointed summits, rising from the seafloor that does not reach the surface of the ocean. Seamounts are volcanic in origin. These can be 3,000-4,500 m tall. The Emperor seamount, an extension of the Hawaiian Islands in the Pacific Ocean, is a good example

Guyots

It is a flat topped seamount. They show evidence of gradual subsidence through stages to become flat topped submerged mountains. It is estimated that more than 10,000 seamounts and guyots exist in the Pacific Ocean alone

Trenches

These areas are the deepest parts of the oceans. The trenches are relatively steep sided, narrow basins. They are some 3-5 km deeper than the surrounding ocean floor. They occur at the bases of continental slopes and along island arcs and are associated with active volcanoes and strong earthquakes. That is why they are very significant in the study of plate movements. As many as 57 deeps have been explored so far; of which 32 are in the Pacific Ocean; 19 in the Atlantic Ocean and 6 in the Indian Ocean

Q.13) Ans: C

Exp:

- Ocean currents can also be classified based on temperature : as cold currents and warm currents.
- **Statement 1 is correct:** cold currents bring cold water into warm water areas. These currents are usually found on the west coast of the continents in the low and middle latitudes (true in both hemispheres) and on the east

coast in the higher latitudes in the Northern Hemisphere

- **Statement 2 is correct:** Warm currents bring warm water into cold water areas and are usually observed on the east coast of continents in the low and middle latitudes (true in both hemispheres). In the northern hemisphere they are found on the west coasts of continents in high latitudes.

Source:

<https://ncert.nic.in/ncerts/l/kegy214.pdf>

Q.14) Ans: D

Exp:

Ocean bottom floor is an outcome of volcanic, tectonic, erosional, depositional process and their interaction. Hence option D is correct.

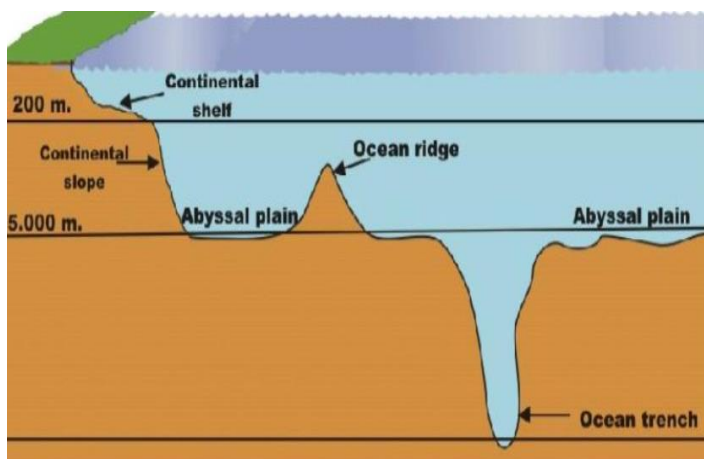
Source: 11th class NCERT

Q.15) Ans: A

Exp:

Statement 1 is correct: A continental shelf is a portion of a continent that is submerged under an area of relatively shallow water known as a shelf sea. Much of these shelves has been exposed during glacial periods and interglacial periods. It is the shallowest part of the ocean showing an average gradient of 1° or even less.

- **Statement 2 is incorrect:** The width of the continental shelves varies from one ocean to another. The average width of continental shelves is about 80 km.



therefore the annual insolation received by the earth on 3rd january is slightly more than the amount on 4th july. However, this variation in the solar output does not have a noted effect on daily weather changes on the surface of the earth.

Point 3 is correct: The capacity of a surface to reflect the sun's energy is called albedo. High albedo has a high percentage of reflection (snow and ice are good reflectors). Temperature will be higher at a given location if its surface has a low albedo rather than a high albedo.

Similarly when the diameter of the dust particles or the vapour droplets is larger than the wavelengths of the incoming solar radiation, some portion of the incoming solar radiation is reflected back to space, this is called diffused reflection.

Point 4 and 5 are correct: Tropical and subtropical areas having vertical rays of the sun and longer length of the day receive maximum insolation. On the other hand, middle and high latitude areas get less insolation, due to oblique angle of the sun's rays and shorter duration of the day. Oblique rays lose more energy, through the mechanism of scattering, reflection, absorption, as they traverse larger distance than vertical rays.

Q.16) Ans: D

Exp:

- A is incorrect: Ridges are formed along a divergent boundary not convergent boundary.
- B is incorrect. Trenches are formed along a convergent boundary not divergent boundary.
- C is incorrect: Canyons are a result of erosional landform.

Q.17) Ans: D

Exp:

- **Statement 1 is incorrect:** A horizontal movement of water often accompanies the rising and falling of the tide. This is called the tidal current. The incoming tide along the coast and into the bays and Estuaries are called a flood current.
- **Statement 2 is incorrect:** The outgoing tide is called an ebb current. The strongest flood and ebb currents usually occur before or near the time of the high and low tides

Q.18) Ans: D

Exp:

Point 1 is correct: Before reaching the earth's surface about 14% of the incoming solar radiation is absorbed mainly by ozone followed by water vapour, carbon dioxide, etc. Presence of clouds, dust particles in the atmosphere affect the distribution of insolation on the earth's surface.

Point 2 is correct: Due to revolution of the earth, the earth is farthest from the Sun on 4th july(aphelion) and nearest to the earth on 3rd january(perihelion),

Q.19) Ans: A

Exp:

Besides major wind systems of the earth's surface, there are certain types of winds, even on a much smaller scale, which are produced by the topographical peculiarities or local temperature differences. Since these wind systems are generated by purely local factors and their zone of influence is quite limited, they are termed as local winds. They play an important role in the weather and climate of a particular locality. Their nomenclature is derived from the regional language. Some of the important local winds with their nature and region of occurrence:

Names: Nature: Region:

Fohn	Warm	Alps.
Chinook	Warm	Rockies.
Berg	Warm	South Africa.
Zonda	Warm	Andes.
Santa Ana	Warm	Coastal southern California.
Khamsin	Warm	Egypt.
Harmattan	Warm	Guinea coast.

Mistral	Cold	South East France.
Samun	Warm	Iran.
Purga	Cold	Russia.
Levanter	Cold	France.
Pampero	Cold	South America.

Q.20) Ans: B

Exp:

Cyclones, hurricanes and typhoons, differing only in name, transfer energies equal to several megaton nuclear bombs, from the ocean to the atmosphere every year. The Coriolis force, the deflecting force caused by the rotation of the earth about its own axis, is maximum at the poles and progressively decreases to zero at the equator. It is this force that compels the surface winds to spiral towards the low pressure system. Tropical cyclones are the most destructive and violent type of storms.

Cyclones:	Regions:
Typhoons	China sea.
Tropical cyclones	Indian ocean.
Hurricanes	Caribbean Sea.
Tornadoes	USA,
Willy Willies	Northern Australia.
Baguio	Philippines.
Taifu	Japan.

Q.21) Ans: d

Exp:

Statement 1 is correct: The position of the jet stream denotes the location of the strongest temperature contrasts between the different latitudes on the Earth surface. The jet stream results from the horizontal differences in temperature that occurs through a substantial distance in the troposphere. The resulting temperature differences produce a horizontal pressure gradient that drives geostrophic and gradient winds. They are marked by strong temperature gradients in the troposphere rather than surface temperature contrasts.

Statement 2 is correct: Jet Streams play an important role in determining local weather phenomena due to their ability to separate colder air and warmer air. The vertical movements of the warmer jet streams are more pronounced and as a result bring varied types of weather conditions to the area. Jet streams generally

push air masses along with them, thus influencing the local weather conditions. For example, when the jet stream extends southwards in the winter (equatorward), it takes the cold polar air masses with it.

Statement 3 is correct: It plays an important role in the onset, development and withdrawal of Indian monsoons. Onset of south-west monsoons in south Asia is because of the withdrawal of the southern branch of the subtropical jet stream to north of the Himalayas, due to the heating of land and atmosphere in summer. Similarly the eastern jet stream enhances the rainfall in south-east Asia and Arabian sea as summer heating causes development of easterly jet streams around 25 degree North latitude.

Statement 4 is correct: Jet streams carry pollutants and other wastes over great distances with a rapid speed. It transports a wide range of pollutants ranging from radioactive and volcanic dust to industrial and other common pollutants. Pilots also fly eastward to take advantage of the jet stream and reduce travel time and fuel usage.

Q.22) Ans: b

Exp:

Pair 1 is not correctly matched: First stage is characterized by convergence of two air masses of contrasting physical properties and directions. It is also known as the initial stage when the warm and cold air masses move parallel to each other and a stationary front is formed.

Pair 2 is not correctly matched: Second stage is known as the incipient stage and during this stage the warm and cold air masses penetrate into the territories of each other and a wave-like front is formed.

Pair 3 is correctly matched: Third stage is also known as the mature stage, because the cyclone is fully developed and the isobars appear to have a circular shape.

Pair 4 is correctly matched: Fourth stage is characterized by narrowing of the warm sector due to progression due to the cold front coming nearer to the warm front.

Pair 5 is not correctly matched: Fifth stage starts with the occlusion of a cyclone when the advancing cold front overtakes a warm front and an occluded front is formed.

Pair 6 is not correctly matched: In the sixth stage the warm sector disappears completely and the cyclone dies out due to the destruction of the occluded front, This stage is also known as frontolysis.

Q.23) Ans: D

Exp:

Statement 1 is correct: Fronts are most clearly developed in the middle latitudes where two air masses, the polar cold air mass and the tropical warm air mass converge. They are characterized by steep gradient in temperature and pressure. They are also responsible for bringing abrupt changes in temperature and cause the air to rise to form clouds and cause precipitation.

Statement 2 is correct: The clearly defined boundary between a warm and a cold air mass, where the latter is advancing and undercutting the warm air. The gradient of the cold front is usually much steeper than that of a warm front. The rapid ascent of warm moist air at the front results in rapid cooling, condensation and the development of cumulonimbus clouds.

Statement 3 is correct: Cold fronts move faster than warm fronts because cold air is denser, meaning there are more molecules of material in cold air than in warm air. Strong, powerful cold fronts often take over warm air that might be nearly motionless in the atmosphere. As the cold front normally travels more quickly than the warm front, it slowly reduces the area of the warm sector, until it merges with the preceding front to complete the occlusion process.

Statement 4 is correct: On weather maps, cold fronts are illustrated by blue lines with sharp "teeth" pointing in the direction of the wind. Warm fronts are illustrated by red lines with rounded bumps pointing in the direction of the wind.

Q.24) Ans: D

Exp:

Statement 1 is correct: Anticyclones is a high-pressure area surrounded by low pressure areas on all sides.

Wind blows from the centre to the outward regions. The rotation of the earth and the blowing of winds from the centre causes the anticyclones to move in clockwise direction in the northern hemisphere and anti-clockwise direction in the southern hemisphere. The isobars in an anticyclone are more or less circular in shape.

Statement 2 is correct: High pressure systems are generally associated with light winds at the surface and subsidence of air from higher portions of the troposphere. Subsidence will generally warm an air mass by adiabatic heating. Thus, resulting in clear skies and rainless because the descending winds cause atmospheric stability.

Statement 3 is correct: Temperature in anticyclones depends on weather, nature of air mass and humidity in the air. They record high temperatures during the summer season due to development of warm air masses.

Q.25) Ans: C

Exp:

Statement 1 is correct: The amount and frequency of rain decreases towards the end of the rainy season and the retreating monsoon are marked by clear skies and rise in temperature, dry season returns in October, with the out blowing North East monsoon. This phenomenon is also known as October heat. In the second half of October the temperature drops rapidly.

Statement 2 is correct: Retreating monsoon is characterized by retreating gradually southwards after mid-september until it leaves the sub-continent completely. The Punjab plains which receive the south-west monsoon earliest are the first to see the withdrawal of the monsoon.

Q.26) Ans: C

Exp:

- **Statement 1 is incorrect:** The heating of the Earth's atmosphere occurs **indirectly** by the **terrestrial radiation** (reflected radiation from the Earth's surface).
- **Statement 2 is correct:** The heating of the Earth's atmosphere through terrestrial radiation is responsible for the decrease in

temperature as one moves up the surface of Earth. Hence, as the distance from the sea level increases, there occurs a fall in temperature. This fall in temperature is defined as the **normal lapse rate**. The areas situated at the higher altitudes experience lower temperature due to this. The normal lapse rate is 6.5°C fall in the temperature with every 1 km increase in height.

- **Statement 3 is correct:** Since land gets heated up quickly and cools down faster compared to the oceans. The land areas are subjected to greater temperature changes. Sudden increase or decrease in the temperature of land is quickly reflected.

Q.27) Ans: B

Exp:

- **Statement 1 is incorrect:** Most of the area of the Southern Hemisphere is covered by the oceans and there is lesser area covered by land. Hence, isotherms (line joining the places with same temperature) are more parallel and show lesser deviation as compared to the Northern Hemisphere.
- **Statement 2 is correct:** During winters in the Northern Hemisphere, the landmass in the North gets cooled quickly and hence isotherms show a bend towards the south while it bends towards the north in the oceans. The most profound southward bending is visible in Siberian region.

Q.28) Ans: A

Exp:

- **Statement 1 is correct:** Beneath the layer of temperature inversion, there occurs a collection of dust particles and smoke which causes dense fogs in the morning during the winter seasons.
- **Statement 2 is correct:** Air drainage is visible along the mountains and hills where the cold air being denser flows down the mountains and accumulates in the bottom areas whereas the area in the upper atmosphere is warmer.
- **Statement 3 is incorrect:** As the Earth's atmosphere is heated indirectly by the

terrestrial radiation of the Earth, the increase in distance from the sea level causes reduced temperature conditions as defined by the normal lapse rate (fall of 6.5 C with every 1 km increase in height).

- **Statement 4 is correct:** On a winter night with clear sky the condition of temperature inversion is best developed as the heat is radiated by the Earth's surface and the lower layers of the atmosphere gets warmer while the Earth's surface is cooler.

Q.29) Ans: D

Exp:

- **Statement 1 is incorrect:** Isobars are the lines joining the areas of same pressure. The changes in Pressure with respect to distance is referred to as the **Pressure Gradient Force**. In cases where the isobars are very close together, the pressure gradient force is strong whereas it is weak in the areas where isobars are far apart.
- **Statement 2 is incorrect:** Frictional force is acting closer to the surface. Its effect reduces beyond the distance of 1-3 km from the surface of the Earth. It has a greater effect over the land surface as compared to the sea surface.
- **Statement 3 is incorrect:** The Coriolis force exists on account of the **rotation of the Earth**. In the Southern Hemisphere, this force deflects the wind towards the left while it deflects them over right in the Northern Hemisphere. Greater the speed of the wind, more is the Coriolis force acting on it.
- **Statement 4 is correct:** The magnitude of Coriolis force is directly proportional to the angle of latitude. Near the Equator, it is practically zero and hence Coriolis force is absent at the Equator while it is maximum at the poles.

Q.30) Ans: A

Exp:

- **Statement 1 is correct:** The upper atmosphere is rarefied in nature and this causes absence of

frictional force. In the absence of force of friction, the Pressure Gradient Force and the Coriolis Force are the sole forces acting on the air. The effect of these two forces is such that the resulting air moving from high pressure to low pressure area is deflected in the direction **perpendicular** to the Pressure Gradient Force. This wind is termed as the **Geostrophic wind**. It bends in the clockwise direction in the Northern Hemisphere.

- **Statement 2 is incorrect:** Besides the Pressure Gradient Force, Coriolis Force, and the Frictional Force, there also exists gravitational force acting in the downward direction.

Q.31) Ans: C

Exp:

- **Statement C is correct:** The factors that are responsible for the distribution of planetary winds are:
 - The **rotation** of the Earth.
 - **Latitudinal** variations in atmospheric heating.
 - **Land and ocean configurations** of the surface of the Earth.
 - Emergence of **pressure belts**.
 - The **oscillation of pressure belts** in accordance with the position of the sun.

The factors like the gravitational pull of the moon and the sunspot cycle are not known to affect the distribution of planetary winds.

Q.32) Ans: C

Exp:

- **Statement 1 is correct:** Trade winds blow on both the sides of the Equator. They blow towards the **Equatorial low pressure**. The westerlies blow at the higher latitudes. They blow from the subtropical high to the polar low-pressure belts.
- **Statement 2 is incorrect:** **Inter-Tropical Convergence Zone** is created at the point where the North and the South East trade

winds converge near the Equator. It is an area of low pressure.

- **Statement 3 is correct:** The air flowing around the Equator are the trade winds which are **Easterlies** in nature. They originate from the North East and South East direction.

Q.33) Ans: D

Exp:

- **Statement 1 is incorrect:** **Hadley cell** is created due to the circulation of air from the Equatorial low-pressure belt to Subtropical high-pressure belt. It is located in the tropics.
- **Statement 2 is incorrect:** **Ferrel cells** are created due to the circulation of air from the Subtropical high to the subpolar low-pressure belt. Such a cell is created near the subtropical high-pressure areas.
- **Statement 3 is incorrect:** The cold air from the poles descends and flows towards the subpolar low-pressure belt. Hence, the polar cell is created around the subpolar low-pressure belts.

Q.34) Ans: A

Exp:

- **Statement 1 is correct:** The general air circulation of the atmosphere also drives the circulation of ocean waters and hence plays an important role in circulating the heat to the entire surface of the Earth. It maintains the circulation of cold air from high to low latitudes and vice versa thereby creating a balanced condition.
- **Statement 2 is incorrect:** The interaction between the ocean and the air in the atmosphere is not unidirectional. The oceans also play a role in the transfer of energy and water vapor to the air mass.

Q.35) Ans: A

Exp:

- **Statement 1 is correct:** El Nino is associated with the warming and cooling of the waters of the Pacific Ocean. The warm pacific water moves towards the Peruvian coast of South America and replaces its cold coastal waters with the warm waters off the coast of Peru. This event is referred to as **El Nino**.
- **Statement 2 is correct:** Northern movement of sun to the Tropic of Cancer in summers is responsible for the heating of the Tibetan Plateau and creation of Low pressure which causes the air to rise above. This creates a strong Tropical Jet Stream over Tibet which is temporary. The stronger the Tropical Jet Streams the more pronounced is the Indian Monsoon.
- **Statement 3 is incorrect:** The El Nino event is very closely related to the event of **Southern Oscillations**. It is associated with changes around the Central Pacific Ocean and Australia. The combined effect of El Nino and southern Oscillation (ENSO) is responsible for causing rains in South America, floods in China, and droughts or lack of monsoon in Australia and India.

Q.36) Ans: D

Exp:

- **Statement 1 is incorrect:** Land has a property of getting heated and cooled faster as compared to sea. Hence, during the daytime, the land gets quickly heated up and creates a low pressure whereas the sea has high pressure. Therefore, the air tends to blow from the sea to land as the pressure gradient exists from sea to land. Whereas on the other hand, during the night, the land gets cooled very quickly causing the air to flow from land to sea.
- **Statement 2 is incorrect:** **Katabatic winds** are cold dense air which flow down the valleys or mountains during the night. The moisture laden winds when crossing a mountain cause condensation and precipitation. On the leeward side these winds are dry and while descending they are warmed up by the adiabatic process.

Q.37) Ans: B

Exp:

- **Statement 1 is incorrect:** Fronts are formed when two air masses meet each other. At the zone of their meeting fronts are formed. Fronts could be either cold or warm. Warm fronts are formed when a warm air mass pushes a cold air mass and causes the upliftment of the warm air mass. A cold front is formed when a cold air mass pushes and uplifts the warm air mass. Fronts are formed in the mid latitudes.
- **Statement 2 is correct:** At the fronts, there occurs sharp changes in temperature and precipitation. The rising warm air forms clouds and causes precipitation.
- **Statement 3 is incorrect:** Occluded fronts are formed when two cold air masses come together and push warm air masses in between them causing rising air and precipitation. Hence, in case of an occluded front, the air mass is lifted fully above the ground.

Q.38) Ans: D

Exp:

- **Statement A is incorrect:** The extra tropical cyclones are of **frontal origin** and are **dynamically induced**. They are formed in the mid latitudes when different air masses meet each other. They move in the direction from **west to east**.
- **Statement B is incorrect:** At the initial stage, there is a formation of **stationary fronts**. Occluded fronts are formed not initially but towards a later stage.
- **Statement C is incorrect:** The extra tropical cyclones are known to cover a greater geographical area and are less destructive compared to the tropical cyclones.
- **Statement D is correct:** The extra tropical cyclones owing to their frontal origin can originate either on the land or on the sea.

Q.39) Ans: B

Exp:

- **Statement 1 is incorrect:** The tropical cyclones are **thermally induced low pressure** systems. They travel in the **East to west direction**.
- **Statement 2 is correct:** The essential prerequisite for the formation of the tropical cyclones is the large sea surface with a temperature greater than 27°C. They are not formed on land and on reaching the land surface, they die out and lose their energy.
- **Statement 3 is incorrect:** A prior low-pressure air or a weak pre-existing cyclonic system is required for their formation. They also require the upper divergence above the sea system.
- **Statement 4 is incorrect:** Tropical cyclones cannot be formed at the Equator as Coriolis force is zero at the Equator. Hence, presence of Coriolis force is another condition for their formation.

Q.40) Ans: C

Exp:

- **Statement 1 is incorrect:** Frequency and destructivity of cyclones are greater in the Bay of Bengal as against the Arabian Sea.
- **Statement 2 is correct:** Since most of the rivers are east flowing in the peninsular part. There is a continuous mixing of the freshwater of the rivers with that of the ocean water in the Bay of Bengal region and hence mixing with the cold bottom water is prevented to some extent. This keeps the surface water warmer for the cyclone formation.
- **Statement 3 is correct:** **Accumulated Cyclone Energy (ACE)** is the measure of the destructiveness of a cyclone. Two or more cyclones occurring together have greater ACE and are more destructive. Events of simultaneous cyclones were witnessed in the Arabian Sea recently.

Q.41) Ans: C

Exp:

- **Statement 1 is correct:** **World Meteorological Organization** and the **United Nations Economic and Social Commission for Asia and Pacific** are responsible for initiating a system for the naming of the Cyclone. As per this

system, eight countries (Bangladesh, India, Maldives, Myanmar, Thailand, Sri Lanka, Pakistan, and Oman) around the Indian Ocean give a list of names which are used for naming of the cyclones.

- **Statement 2 is correct:** Arabian Sea is considered as a closed sea and is hence favourable for the formation of cyclones. Bay of Bengal is regarded as an open sea.

Q.42) Ans: B

Exp:

- **Statement 1 is incorrect:** Fronts are not formed at the lower latitudes. They are formed only in the mid latitudes. Hence, warm fronts cannot be formed around the Equatorial forests.
- **Statement 2 is correct:** The center of a cyclonic system contains a region of intense calm and subsiding air. This is a narrow high-pressure area. This area is often confused with the end of the cyclone while the other half still remains.

Q.43) Ans: A

Exp:

- **Statement 1 is correct:** The **anticyclonic pressure systems** are characterized by winds moving clockwise in the Northern Hemisphere and anticlockwise in the southern Hemisphere. The **cyclonic pressure systems** are characterized by anticlockwise winds in the Northern Hemisphere and clockwise winds in the Southern Hemisphere.
- **Statement 2 is incorrect:** The vertical pressure gradient is always more pronounced as compared to the horizontal pressure gradient. However, the vertical pressure gradient is balanced by the presence of gravity.

Q.44) Ans: C

Exp:

- **Statement 1 is correct:** The moon's gravitational pull to a great extent and to a lesser extent, the sun's gravitational pull are the main reasons behind tides. Another factor is the centrifugal force



which acts to counter the gravitational pull, and the two together are responsible for creating two major tidal bulges.

- **Statement 2 is correct:** Movement of water caused by meteorological effects are called Surges. They are different from tides because they do not originate from gravitational pull. They are the result of winds and pressure changes in the atmosphere. They are not regular like tides.

Source: NCERT

Q.45) Ans: D

Exp:

Latitude - The temperature of ocean water decreases from the equator towards the poles because the amount of insolation decreases as we move poleward. The oceans in the northern hemisphere record relatively higher temperatures than in the southern hemisphere because of the concentration of landmass in the north. The highest temperature is recorded not at the equator but slightly north of it.

Land-water distribution - In the northern hemisphere the oceans receive more heat due to their contact with a larger extent of land than the oceans in the southern hemisphere. As land heats faster and cools down faster than water, the surrounding waters are impacted accordingly.

Prevailing winds - The winds blowing from the land towards the oceans drive warm surface water away from the coast resulting in the upwelling of cold water from below. It results in longitudinal variation in the temperature. Contrary to these onshore winds pile up warm water near the coast and this raises temperature.

Ocean currents - Warm Ocean currents raise the temperature in cold areas while cold currents decrease the temperature in warm ocean waters. The Gulf stream ensures countries of western Europe have access to warm water ports throughout the year. While the oyashio current freezes the waters in Eastern Russia.

Source: Geography Ncert

UPSC Q referred: Pre 2013 world geography Q

Q.46) Ans: A

Exp:

Statement 1 is correct - There is a boundary layer of water between the surface and the deeper layers. This boundary layer of water witnesses rapid decrease of temperature and is called THERMOCLINE. The boundary begins around 100-400m below the sea surface and extends several meters down. The rate of decrease of temperature with increasing latitude is 0.5 degrees per latitude. The avg temperature of surface waters is around 27 degrees.

Statement 2 is Incorrect - At the equator there is strong evaporation, cloud formation and rain which brings the temperature down. Plus, there is upwelling due to evaporation which brings colder sea water to the surface from the deep. Hence tropical waters with little cloud cover and upwelling are warmer than equatorial waters. (Understand equatorial as between 0 to 10 degrees N/S while tropical as between the two tropics.)

Source: Geography Ncert

UPSC Q referred: Pre 2018 Q 55 set D

Q.47) Ans: D

Exp:

Statement 1 is Correct: Salinity is measured in "parts per thousand". It is calculated as the amount of salt dissolved in 1000gm of seawater. The salinity of water in the surface layers of the ocean depend mainly on evaporation and precipitation. As the water evaporates or turns into ice, the salt particles remain free of water molecules and thus the salinity increases.

Statement 2 is Correct: The ocean currents contribute to the salinity variations. Salinity, temperature and density of water are interrelated. Hence, Any change in the temperature or density influences the salinity of water in an area. Salinity usually increases with density ie., there is a positive correlation between the two. In a zone called HALOCLINE, salinity increases very sharply. The denser layers of water sink and this means that with depth, salinity will increase. This leads to stratification by salinity.

Source: NCERT geography



UPSC Q referred: Pre 2020 set B Q 53

Q.48) Ans: D

Exp:

Factors Affecting Ocean Salinity

Evaporation and precipitation - Evaporation and precipitation simultaneously adds or reduces the salt content in the oceans thus increasing or reducing the salinity. Due to evaporation, salt particles are left behind in the ocean and thus salinity increases. While precipitation increases the amount of freshwater, thus reducing salinity.

Freezing and thawing of ice - Freezing and thawing of ice simultaneously increases and reduces the salt content in the liquid water, thus affecting salinity. When ocean water freezes, the salt particles remain free of ice and the water that is frozen is mostly freshwater. Freezing increases the salt content while thawing adds freshwater and reduces salinity.

Inflow of fresh water - It reduces the salinity as is witnessed in the case of Black sea (Volga, Dnieper etc drain into it) and Bay of Bengal. These water bodies otherwise would have been high in salinity due to being enclosed and hence not having intermixing of water layers and masses.

Wind - Wind influences salinity of an area by transferring water to other ocean zones and helping in intermixing. The wind also impacts salinity indirectly by ocean currents, swells, precipitation etc.

Ocean Current - Indirectly, the winds through the ocean current determine the inflow and outflow of ocean water with different salinity and hence intermixing of water masses, thus impacting salinity.

UPSC Q referred: Pre 2013 world geography Q

Q.49) Ans: C

Exp:

Pair 1 is Incorrect - Semi-diurnal tide is the most common tidal pattern where there are two high tides and two low tides each day. The interval is not exactly 12 hrs; instead, that of 12 hrs 25 minutes. These are

observed in most areas and are also called semi-day tides. In rare cases there can be 3-4 tides.

Pair 2 is Incorrect - Diurnal tide is when there is only one high tide and one low tide each day. Eg is the tide experienced in the areas around the Gulf of Mexico.

Pair 3 is Correct - Spring tide is when the sun, moon and earth are in a straight line. The height of the tides are higher. Spring tides happen whenever there is a new moon or a full moon and have nothing to do with the season of spring.

Pair 4 is correct - Neap tide is when the sun, the moon and the earth are aligned in right angles such that the pull of both the heavenly bodies cancel each other out and tides are weak. In the period between the two spring tides, the moon faces the Earth at a right angle to the sun. When this happens, the pull of the sun and the moon are weak. This causes tides that are lower than usual. These tides are known as neap tides.

Extra Edge by Only IAS

Information about

similar events

Ebb tide: One high tide always faces the moon, while the other faces away from it. Between these high tides are areas of lower water levels—low tides. The flow of water from high tide to low tide is called an ebb tide.

Tidal Bore: A tidal bore occurs along a coast where a river empties into the ocean or sea. The tidal bore is a strong tide that pushes up the river, against the river's current. It is called Pororoca in the Amazon river.

Rip tide: Rip tides are strong ocean currents running along the surface of the water. A rip tide runs from the shore back to the open ocean. They can be very dangerous to swimmers, who can be swept out to sea. They are not tidal features.

Source: Ncert, NOAA website, National Geographic

Q.50) Ans: A

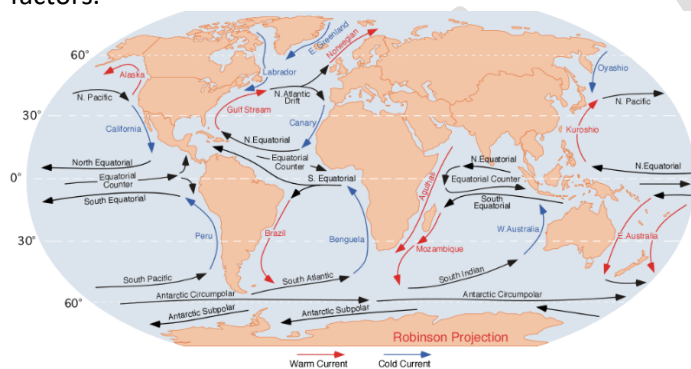
Exp:

Statement 1 is correct - There is a strong correlation in the patterns of winds and the movement of ocean water. Major ocean currents are greatly influenced by stresses exerted by prevailing winds and coriolis force. The air circulation in the mid-latitudes is mainly anticyclonic and in the lower latitudes it's cyclonic. The

oceanic circulation pattern also corresponds with the same.

Statement 2 is correct - The Current's speed is called "drift" and it is measured in terms of knots. A current is usually strongest at the surface and decreases in strength with depth. At a surface a current may attain speeds over 5 knots whereas at depths speed is less than 0.5 knots. This is due to increased exposure of the surface layer to the winds. Especially in the southern hemisphere beyond 40 degree latitude, the impact of winds can be clearly observed.

Statement 3 is incorrect - COLD CURRENTS bring cold water into warm water areas. These currents are usually found on the west coast of continents in the low & middle latitudes and on the east coast of higher latitudes. They impact the formation and expansion of deserts on each continent, which are mostly found on the western coast of continents. On The Atlas it can be observed that major cold currents are located near the major deserts of the world. This is due to the dryness of air in the corresponding areas along with other factors.



Extra Edge by Only IAS

- Currents based on temperature can be classified into - warm currents & Cold Currents
- Warm Currents bring warm water into cold water areas and are usually found on the east coast of continents in the low and middle latitudes and west coast of continents in higher latitudes.

Q.51) Ans: C

Exp:

Statements 1 & 2 are incorrect

Statement 3 is Correct - The atmosphere is composed of gases, water vapour and dust particles. In addition, it contains solid and liquid particles, collectively called

'aerosols in large numbers'. The air is an integral part of the earth's mass & 99 % of the mass of the atmosphere is confined to the height of 32 Km from the earth surface.

Statement 4 is Incorrect - The proportion of gases are not the same throughout all the layers of atmosphere. It changes in the higher layers of the atmosphere. At the height of 120 km the oxygen is found in almost negligible quantity. Water vapour and carbon dioxide are also found only up to 90 km from the surface of the earth.

Extra Edge by Only IAS

- The atmosphere of Earth is composed of nitrogen (78.08%), oxygen (20.95%), argon (0.93%), carbon dioxide (0.036%) and other gases such as neon, helium, hydrogen etc.
- Water vapours decrease with altitude. They also decrease from the equator towards the poles. It also absorbs parts of the insolation from the sun and preserves the earth's radiated heat. Thus water vapours act like a blanket allowing the earth neither to become too cold nor too hot.
- Dust particles are generally concentrated in the lower layers of the atmosphere. **The higher concentration** of dust particles is found in subtropical and temperate **regions due to dry winds**.

Source: Ncert- Class 11th Geography

Q.52) Ans: D

Exp:

Statement 1 is Incorrect - Carbon dioxide is **transparent** to the incoming solar radiation but **opaque** to the outgoing terrestrial radiation. It absorbs a part of terrestrial radiation and reflects back some part of it towards the earth surface. It is largely responsible for the greenhouse effect.

Statement 3 is Incorrect - The volume of other gases is constant but the volume of carbon dioxide has been rising in the past few decades mainly due to the burning of fossil fuels. It has also increased the temperature of the air.

Source: NCERT- class 11th

Q.53) Ans: D

Exp:

Statement 1 is incorrect: Nitrogen and oxygen make up nearly 99% of the clean, dry air. The remaining gases are mostly **inert** and constitute about 1% of the atmosphere

Constituent	Formula	Percentage by Volume
Nitrogen	N ₂	78.08
Oxygen	O ₂	20.95
Argon	Ar	0.93
Carbon dioxide	CO ₂	0.036
Neon	Ne	0.002
Helium	He	0.0005
Krypto	Kr	0.001
Xenon	Xe	0.00009
Hydrogen	H ₂	0.00005

Statement 2 is incorrect- Ozone is an important component of the atmosphere found between 10 and 50 km above the earth's surface and acts as a filter and absorbs the **ultraviolet rays** (not infrared rays) radiating from the sun and prevents them from reaching the surface of the earth.

Q.54) Ans: D

Exp:

Statement 1 is Incorrect - Water vapours are not fixed in the atmosphere. They decrease with altitude. In the warm and wet tropics, it accounts for about 4 % of the air by volume, whereas in the dry and cold areas of desert and polar regions, it may be less than 1 % of the air. Water vapour decreases from the equator towards the poles.

Statement 2 is Incorrect - Dust particles are generally concentrated in the lower layers of the atmosphere; yet, convection air currents may transport them to great heights.

Statement 3 is correct - The higher concentration of dust particles is found in subtropical and temperate regions due to dry winds as compared to the equatorial and polar regions. Dust and salt particles act as

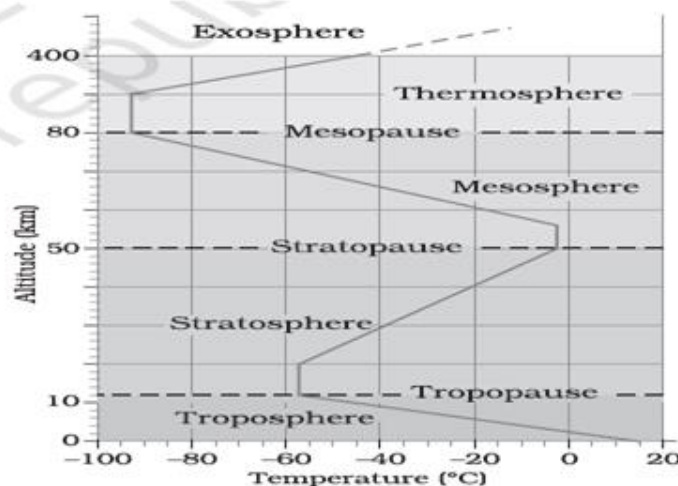
hygroscopic nuclei around which water vapour condenses to produce clouds.

Q.55) Ans: D

Exp:

Statement 1 is Incorrect - The atmosphere consists of different layers with varying density and temperature. Density is highest near the surface of the earth and decreases with increasing altitude.

Statement 2 is Incorrect - The column of atmosphere is divided into five different layers depending upon the temperature condition - **Troposphere, Stratosphere, Mesosphere, Thermosphere and Exosphere**. The troposphere is the lowermost layer and the exosphere is the highest layer of the atmosphere.



Q.56) Ans: B

Exp:

Statement 1 is Incorrect - The troposphere is the lowermost layer of the atmosphere. The average height of this layer is 13 km. It extends broadly to a height of 8 km near the poles & about 18 km at the equator. Thickness of the troposphere is greatest at the equator because heat is transported to great heights by strong convective currents. The temperature in the troposphere decreases at the rate of 1°C for every 165m of height, the fall in temperature is called 'lapse rate'.

Statement 2 is correct - Troposphere contains dust particles and water vapour. All the changes in climate

and weather take place in this layer. Troposphere is the most important layer for all biological activity. Tropopause is the zone which separates the troposphere from the stratosphere. The temperature here is nearly **constant, hence, it is called the tropopause.**

Source: NCERT – class 11th

Q.57) Ans: B

Exp:

Statement 1 is Correct - In Thermosphere temperature increases with height. It is located between 80 and 400 km above the mesopause. It contains electrically charged particles known as ions, and hence, it is known as ionosphere. Radio waves transmitted from the earth are reflected back to the earth by this layer. The International Space Station and satellites orbit in this layer.

Statement 2 is Correct - The stratosphere is located above the troposphere and below the mesosphere. In this layer temperature increases with altitude. It is the result of the absorption of the Sun's ultraviolet radiation by the ozone layer. Ozone layer reflects the harmful ultraviolet radiation. The ozonosphere is also called a chemosphere because a lot of chemical activity goes on here.

Statement 3 is Incorrect - The **troposphere** is the lowermost layer of the atmosphere. The average height of this layer is 13 km. It extends broadly to a height of 8 km near the poles & about 18 km at the equator. The temperature in the troposphere decreases at the rate of 1°C for every 165m of height, the fall in temperature is called 'lapse rate'.

Statement 4 is Incorrect - **Mesosphere** lies above the stratosphere, which extends upto a height of 80 Km. In the Mesosphere temperature starts decreasing with the increase in altitude and reaches up to minus 100°C at the height of 80 km. Mesopause is the uppermost limit of the Mesosphere.

Q.58) Ans: A

Exp:

Statement 1 is Correct - Exosphere is the uppermost layer of the atmosphere. It gradually merges with outer space. Light gases like helium and hydrogen

float into space from here. Very little is known about it due to lack of research.

Statement 2 is Incorrect - The thermosphere extends from 80-400 km not exosphere. Exosphere extends from 400 Km and lies above the thermosphere.

Q.59) Ans: D

Exp:

Statement 1 is Incorrect - During its revolution around the sun, the earth is farthest from the sun (152 million km) on 4th July. This position of the earth is called *aphelion*.

Statement 2 is Incorrect - On 3rd January, the earth is the nearest to the sun (147 million km). This position is called *perihelion*.

Extra Edge by Only IAS

- The solar output received at the top of the atmosphere varies slightly in a year due to the variations in the distance between the earth and the sun. Therefore, the annual insolation received by the earth on 3rd January is slightly more than the amount received on 4th July.
- However, the effect of this variation in the solar output is masked by other factors like the distribution of land and sea and the atmospheric circulation. Hence, this variation in the solar output does not have a great effect on daily weather changes on the surface of the earth.

Q.60) Ans: D

Exp:

Statement 1 is incorrect - Conduction takes place when two bodies of unequal temperature are in contact with one another, there is a flow of energy from the warmer to cooler body.

Statement 2 is incorrect - The air in contact with the earth rises vertically on heating in the form of currents and further transmits the heat of the atmosphere. This process of vertical heating of the atmosphere is known as *convection*. The convective transfer of energy is confined only to the troposphere

Statement 3 is incorrect - The transfer of heat through horizontal movement of air is called *advection*. Horizontal movement of the air is relatively more important than the vertical movement. In middle latitudes, most of diurnal (day and night) variation in daily weather are caused by advection alone. In tropical regions particularly in northern India during summer season local winds called 'loo' is the outcome of advection process.

Q.61) Ans: D

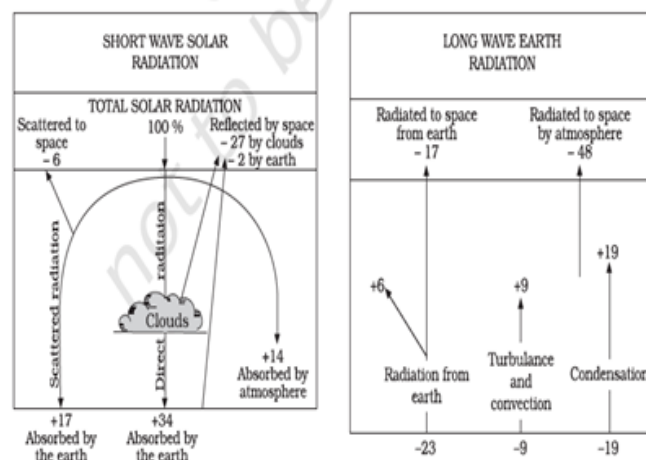
Exp:

Statement 1 is Correct - The earth as a whole does not accumulate or loose heat. It maintains its temperature. This can happen only if the amount of heat received in the form of insolation equals the amount lost by the earth through terrestrial radiation.

Statement 2 is Correct - Heat Budget: Out of the 100 units of incoming solar radiation, 35 units (27-clouds, 2-snow, 6-atmosphere top) are reflected back as albedo. Out of the remaining 65 units, 14 units are absorbed by the atmosphere & 51 units by earth surface.

The earth radiates back 17 units directly into space & the remaining 34 units are absorbed by the atmosphere. The atmosphere then radiates back the 48 units absorbed by it. Thus, the total radiation returning from the earth and the atmosphere respectively is 17+48=65 units which balance the total of 65 units received from the sun.

This is termed as the heat budget or heat balance of the earth. This explains why the earth neither warms up nor cools down despite the huge transfer of heat that takes place.



Heat budget of the earth

Q.62) Ans: A

Exp:

Statement 1 is correct - Germanwatch is an environmental think tank and sustainable development lobbyist. It releases the Global Climate Risk Index. It analyses to what extent countries have been affected by the impacts of weather related loss events. It pushes for the need to support developing countries in coping with the effects of climate change.

Statement 2 is incorrect - India is the 5th most vulnerable country to climate change as per the 2020 report. India has recorded the highest number of fatalities and second highest monetary losses from climate change's impact in 2018.

Statement 3 is incorrect - The "Brown to Green report" is released by "Climate Transparency Partnership" which is an international research collaboration. It is the most comprehensive review of G20's climate performance.

Source: [Download: Global Climate Risk Index 2020](#)

Q.63) Ans: D

Exp:

- Statement 1 is correct:** -In May, the Somali Low-Level Jet (SLLJ) assists the SWM in crossing the Arabian Peninsula and arriving in west India in June. The overall SLLJ course in May

and June is toward India, with wind speeds gradually increasing over time. The wind speed peaks in June and August, resulting in the Northern Hemisphere's highest wind speed.

- **Statement 2 is correct:-** Jet streams are formed by the separation of hot and cold air. Winters in both the northern and southern hemispheres have the strongest jet currents because these hot and cold air boundaries are more pronounced.
- **Statement 3 is correct:-** Recent observations have shown that the strength and length of Tibetan Plateau heating has a direct impact on the amount of rain that the monsoons bring to India. When the summer temperature of the air over Tibet remains high for a long enough period of time, it aids in the strengthening of the easterly jet, resulting in heavy rain in India. If the snow on the Tibetan Plateau does not melt, the easterly jet does not form. Rainfall in India is hampered as a result of this.

Q.64) Ans: C

Exp:

- **Statement 1 is correct:-** Source Regions of its origin The regions where the air masses originate are called source regions. An ideal source region of an air mass should have the following essential conditions:- (i) There should be an extensive and homogeneous surface of the earth so that it may have uniform temperature and moisture conditions. The source region may be entirely a landmass or an entirely an Oceanic surface. (ii) The speed of the wind should be slow and should have divergence, so that The air of other region may not enter. (iii) **The atmospheric conditions should remain constant for considerable longer period of time, the air mass may possess the characteristics of the ground surface.**
- **Statement 2 is incorrect:-** A pre existing Low-level disturbance is essential for formation of tropical cyclone.
- **Statement 3 is correct:-** Source regions are where air masses form over large surfaces with uniform temperatures and humidity. Low wind speeds allow air to remain stationary for long periods of time, allowing it to absorb the characteristics of the source field, such as heat

or cold. Winds transport weather conditions (heat or cold, dry or moist) from one area to another as they pass air masses

Q.65) Ans: D

Exp:

- **Statement 1 is incorrect:-** It exists as a result of the trade winds colliding. The northeast trade winds in the northern hemisphere meet the southeast trade winds in the southern hemisphere. The ITCZ is formed as the trade winds combine and push air up into the atmosphere.
- **Statement 2 is incorrect:-** The ITCZ (Intertropical Convergence Zone) is also known as the Equatorial Convergence Zone or Intertropical Front and plays an important role in the global circulation system. It's a low-pressure belt that encircles Earth near the Equator.
- **Statement 3 is incorrect:-** Mid-latitude depressions, temperate cyclones, frontal depressions, and wave cyclones are also names for extratropical cyclones. In both hemispheres, these are active above the mid-latitudinal area between 35° and 65° latitude. The migration is from west to east, and it is more pronounced during the winter months. The polar and tropical air masses collide in these latitude zones, forming fronts
- **Statement 4 is correct:-** The intertropical convergence zone (ITCZ), also known as the equatorial convergence zone, is a ring of converging trade winds and rising air that circles the globe near the Equator. The rising air causes high cloudiness, regular thunderstorms, and heavy rainfall; the zone also contains the doldrums, oceanic regions of calm surface air. The ITCZ moves north and south with the Sun during the year.

Q.66) Ans: C

Exp:

Statement 1 is incorrect:- El Nino is the warm phase of the El Nino–Southern Oscillation (ENSO), and it's linked to a band of warm ocean water that forms in the central and east-central equatorial Pacific, including off the coast of South America. El Nino causes the cold surface water off the Peruvian coast to warm up. The

regular trade winds get lost or change direction when the water is warm. As a result, moisture-laden winds from the western Pacific are driven towards Peru's coast.

- **Statement 2 is correct:**-Rainfall across Australia could be decreased during El Nino events due to a change in rainfall away from the Western Pacific. As a result of the delayed onset of the Indo-Australian Monsoon in tropical Australia, rainfall in the northern tropics is decreased. Following an El Nino occurrence, the likelihood of a major bushfire season in south-eastern Australia is higher.
- **Statement 3 is incorrect:**El Nino means India will get less rain than normal. Since Indian agriculture is heavily reliant on the monsoons, lower rainfall during the monsoons usually translates to lower crop yields.

Q.67) Ans: A

Exp:

Statement 1 is correct:-The cool temperate western margins are under the permanent influence of westerlies all round the year. The westerlies blow from south-west to north-east in the northern hemisphere and north-west to south-east in the southern hemisphere. They are best developed between 40° and 65°S latitudes.

- **Statement 2 is incorrect:**-About 30° and 45° north and south of the equator, the Mediterranean climate is restricted. The shifting of the wind belts is the primary cause of this form of climate. The belt of power of the Westerlies is moved a little polewards in summer when the sun is overhead at the Tropic of Cancer. As a result, rain-bearing winds are unlikely to hit the Mediterranean. When the Westerlies move equatorwards in the winter, the Mediterranean lands receive the majority of their precipitation.
- **Statement 3 is incorrect:**-Rainfall in Laurentian climate is throughout the year but there is distinct summer maximum from easterly winds from the ocean and winter is dry and cold because the winds are dry westerlies that blow out from continental interior.
- **Statement 4 is incorrect:**-In the winter, an extra tropical storm known as the Western Disturbance brings rain to North and

Northwest India. It's ideal for wheat and other winter crops.

Q.68) Ans: C

Exp:

- **Statement 1 is correct:**-Deformatory circulation involves convergence of two contrasting air masses and horizontal spread along a line which is called as axis of outflow or axis of dialation. This type of circulation is most favourable for frontogenesis.
- **Statement 2 is correct:**-According to polar front theory, Cold, humid air masses from the tropics collide with dry, cold air masses from the poles, forming a polar front as a discontinuous surface.
- **Statement 3 is incorrect:**-Warm front is gently sloping frontal surface along which warm and light become active and aggressive and rises slowly over cold and dense air. Precipitation over such front is in the form of moderate to gentle. On the other hand cold front is associated with much steeper front and characterized by thick cloud, heavy downpour with thunderstorm, lightning etc.
- **Statement 4 is incorrect:**-Contrast in Temperature of two air mass is more in winter session

Q.69) Ans: B

Exp:

- **Statement 1 is incorrect:**- At equator two airmass of relatively same physical characteristic and moisture content converge which is not condition favourable for the formation of temperate cyclone.
- **Statement 2 is correct:**-First stage of Temperate cyclone formation involves the convergence of of two air masses of contrasting physical properties. But at the equator two tropical airmasses meet and both have comparatively same temperature and moisture content.
- **Statement 3 is incorrect:**- Coriolis force at the equator is zero
- **Statement 4 is incorrect:**- Doldrum is located between the latitudes of 10°N and 10°S. This belt is the meeting point of trade winds from

both hemispheres, as well as sub-tropical high pressure belts. Because of the intensely calm air movements, this belt is also known as the Doldrums.

Q.70) Ans: D

Exp:

- **Statement 1 is correct:-** Due to northward migration of sun in northern hemisphere at the time of summer solstice the north Intertropical convergence zone is extended upto 30deg N latitude over Indian subcontinents, South Asia and part of Africa. Thus equatorial westerlies are also extended over the aforesaid region. These equatorial westerlies in fact become the south west monsoon.
- **Statement 2 is correct:-** The subtropical high pressure belts extending between 30°- 35° latitudes are characterized by dry trade winds during summer season. Thus, the western parts of the continents within the zone of 30°-40° latitudes do not receive rainfall during summer season. On the other hand, the subtropical belt shifts towards the equator at the time of winter solstice in the northern hemisphere, the zone between 30°-40° latitude is characterized by westerlies which give much precipitation during winter season because they come from over the oceans.
- **Statement 3 is correct:-** Since the westerlies spread over these areas due to the northward (poleward) shifting of the sub-polar low pressure belt, polar easterlies are weakened during northern summer due to the sun's northward migration at the time of the summer solstice. The situation is reversed at the time of winter solstice when there is southward migration of the sun. Thus, the polar easterlies are re-established over the areas lying between 60°-70° latitudes in the northern hemisphere because of the shifting of the belt of westerlies southward.

Q.71) Ans: A

Exp:

- **Statement 1 is Correct :** Recently, the Ministry of Home Affairs (MHA) has directed Nagaland,

Manipur, Mizoram and Arunachal Pradesh to check illegal influx from Myanmar into India.

- The State governments have no powers to grant "refugee status to any foreigner" and India is not a signatory to the United Nations Refugee Convention of 1951 and its 1967 Protocol.
 - UN Refugee Convention 1951 - : It is a United Nations multilateral treaty that defines who is a refugee, and sets out the rights of individuals who are granted asylum and the responsibilities of nations that grant asylum. It also set out which people do not qualify as refugees, such as war criminals
 - The 1967 Protocol included refugees from all countries as opposed to the 1951 Convention that only included refugees from Europe.
- Similar Instructions were Issued in August 2017 and February 2018.
- **Statement 2 is Correct :** India and Myanmar share a 1,643 km border and people on either side have familial ties.
 - Mizoram shares 510-km.
 - Manipur shares 398-km.
 - Arunachal Pradesh shares 520 kms.
 - Nagaland shares 215 kms

The border along the four states is unfenced and porous.
- **Statement 3 is Incorrect :** A Free Movement Regime (FMR) exists between India and Myanmar.
 - Under FMR every member of the hill tribes, who is either a citizen of India or a citizen of Myanmar and who is resident of any area within 16 km on either side of the Indo-Myanmar Border (IMB) can cross the border with a border pass (with one-year validity) issued by the competent authority and can stay up to two weeks per visit.

Q.72) Ans: C

Exp:

- **Satkosia Tiger Reserve is located in Odisha.** It was declared as Tiger Reserve in 2007, satkosia tiger Reserve had a population of 12 tigers then, the number reduced to two in 2019.
- **India's first inter-state tiger translocation project** was initiated wherein two big cats, a male from Kanha Tiger Reserve and a female (Sundari) from Bandhavgarh tiger reserves (Madhya Pradesh) were relocated to satkosia Tiger Reserve in Odisha, to shore up the tiger population in the state. The purpose of relocation was to repopulate tigers in reserve areas.
- Recently in the news, because tigress Sundari was relocated back to MP. This was mainly because of severe protests by villagers living on the fringes of the reserve.



Source:

<https://www.hindustantimes.com/india-news/odisha-sends-back-tigress-from-satkosia-reserve-to-madhyapradesh-101616554861730.html>

Q.73) Ans: D

Exp:

- The main ethnic group living in seven ethnic minority states of Burma are the **Karen, Shan, Mon, Chin, Kachin, Rakhine, and Karenni**. Other main groups include the Nagas, who live in north Burma and are estimated to number more than 100,000 constituting another complex family of Tibetan-Burmese language sub-group.
- **Hazaras** are considered to be one of the most oppressed ethnic groups in **Afghanistan**.
- The **Yezidis** are one of many religious minorities in **Iraq**. They follow Yezidism, a religion combining elements from, among others, Christianity and Islam.

- The **Uighurs** are a predominantly Muslim minority group predominantly living in **Xinjiang province of China**. Since the last few decades they have suffered from abuses including persecution and forced detention.
- Recently in the news, 11 coal mine workers of the Hazara community were killed by the Islamic State group in Pakistan.

Source:

<https://www.thehindu.com/news/international/pakistani-shiites-end-protests-hold-funeral-for-11-miners/article33536855.ece>

Q.74) Ans: C

Exp:

- **Statement 1 is correct:** The Suez Canal is an artificial sea-level waterway running north to south across the isthmus of Suez in Egypt opened in 1869. The Suez Canal is now controlled by Egypt. This has a significant impact on Egypt's economy.
- **Statement 2 is not correct:** The Suez Canal connects the Mediterranean Sea and the Red Sea.
- **Statement 3 is not correct:** it is one of the world's most heavily used shipping lanes, carrying over 12% of world trade by volume.
- **Statement 4 is correct:** The canal separates the African continent from Asia. It provides the shortest maritime route between Europe and the land lying around India and the Western Pacific Ocean.
- **Recently in the news,** because a large cargo ship named 'Ever Given' got stuck near the southern end of the Suez Canal due to a mishap caused by bad weather. Huge impact on world trade.

Source

<https://indianexpress.com/article/explained/explained-what-has-caused-a-traffic-jam-in-the-suez-canal-7243543/>

Q.75) Ans: D

Exp:

- **Statement 1 is correct:** Under the new guidelines, the sector will be deregulated and aspects such as prior approvals for surveying,

mapping and building applications based on that have been done away with.

- **Statement 2 is correct:** For Indian entities, there will be complete deregulation with no prior approvals, security clearances and licences for the acquisition and production of geospatial data and geospatial data services, including maps
- The location may be static in the short-term, like the location of a road, an earthquake event, malnutrition among children, or dynamic like a moving vehicle or pedestrian, the spread of an infectious disease. Geospatial data combines location information, attribute information (the characteristics of the object, event, or phenomena concerned), and often also temporal information or the time at which the location and attributes exist.
- Geo-spatial data usually involves information of public interest such as roads, localities, rail lines, water bodies, and public amenities. Geospatial data is data about objects, events, or phenomena that have a location on the surface of the earth. The past decade has seen an increase in the use of geospatial data in daily life with various apps such as food delivery apps like Swiggy or Zomato, e-commerce like Amazon, or even weather apps.
- **Recently in the news,** because the Ministry of Science and Technology released new guidelines for the Geo-spatial sector in India, which deregulates existing protocol and liberalizes the sector to a more competitive field.

Source: <https://www.businesstoday.in/current/economy-politics/government-liberalises-rules-on-geospatial-data-issues-new-guidelines/story/431370.html>

Q.76) Ans: D

Exp:

- **Statement 1 is correct:** Article 243k says, superintendence, direction, and control of the preparation of electoral rolls for, and the conduct of all elections to the Panchayat bodies in the State shall be vested in an SEC.
- **Statement 2 is incorrect:** SEC is to be appointed by the Governor. Subject to the provisions of any law made by the Legislature

of a State, the conditions of services and tenure of the office of the SEC shall be such as the Governor may by rule determine.

- **Statement 3 is correct:** SEC shall not be removed from his/her office except in the manner and on the like grounds as a Judge of a High Court.
- **Statement 4 is incorrect:** SEC vested with the election of Panchayat and Municipalities. Election of state legislation is overseen by the Central Election Commission.
- **Recently in the news,** because the independent person and not serving bureaucrats, should be appointed as state election commissioner.

Q.77) Ans: C

Exp:

- **Statement 1 is correct:** Millets are often referred to as superfood and its production can be seen as an approach for sustainable agriculture and a healthy world. The three major millet crops currently grown in India are jowar, bajra and ragi. Along with that, India grows a rich array of bio-genetically diverse and indigenous varieties of "small millets" like Koda, Kutki, chenna and Sanwa.
- **Statement 2 is correct:** Millets are less expensive and nutritionally superior to wheat and rice. millets are also rich in iron, calcium and magnesium. For example - Ragi is known to have the highest calcium content. Its high iron content can fight high prevalence of anaemia in Indian women of reproductive age and infants.
- **Statement 3 is not correct:** They are also harder and drought resistant crops, which has to do with their short growing season (70-100 days) and lower water requirement (300-500).
- **Statement 4 is correct:** Millets can help tackle lifestyle problems and health challenges such as obesity and diabetes as they are gluten-free and have a low glycemic index (a relative ranking of carbohydrates in foods according to how they affect blood glucose levels). Millets are rich in antioxidants properties.
- **Recently in news:** An International fund for agricultural development (IFAD) supporting an initiative to revive Koda and Kutki millet cultivation, started in the year 2013-14 in

Dindori district of Madhya Pradesh, has given new life to the forgotten crops.

Source:

<https://indianexpress.com/article/india/in-madhya-pradeshs-dindoi-a-un-backed-project-aims-to-put-millet-7249608> (<https://indianexpress.com/article/india/in-madhya-pradeshs-dindoi-a-un-backed-project-aims-to-put-millet-7249608/>)

Q.78) Ans: B

Exp:

- Capital Gains Tax is a tax on growth of value of investments incurred when individuals and corporations sell those investments.
- **The tax doesn't apply to unsold investments**, so stock shares that appreciate every year will not incur capital gains taxes until they are sold, no matter how long they are held.
- Capital gains treatment only applies to "capital assets" such as stocks, bonds, jewelry, coin collections, and real estate property among others. In other words, it applies only to transactions which are capital in nature i.e., result in change of assets or liabilities.
- When an asset is held for a period of 36 months or less, it is termed as a short-term capital asset and above 36 months, is considered as long-term capital asset.
- Recently in the news because the Finance Bill, 2021 proposed an amendment to the regulations relating to Capital Gains Tax (CGT).

Source:

<https://www.hindustantimes.com/budget/union-budget-2021-22-here-are-10-income-tax-changes-announced-101612496255265.html>

Q.79) Ans: A

Exp:

- **Statement 1 is correct:** The bill seeks to amend Medical Termination of Pregnancy Act, 1971 to regulate the conditions under which a pregnancy may be aborted and increases the time period from earlier 20 weeks to 24 weeks, within which abortion may be carried out.

- **Statement 2 is correct:** The Bill allows unmarried women to terminate a pregnancy for the reason of failure of contraceptive method or device. Earlier this provision was only for married women.
- **Statement 3 is incorrect:** Bill will help to reduce maternal mortality and morbidity as it increases access of women to legal and safe abortion service in order to reduce maternal mortality and morbidity caused by unsafe abortion and its complications. The word **drastically** in the 3rd statement is extremeness.
- **Recently in the news because:** Rajya Sabha has passed the Medical Termination of Pregnancy (MTP) (Amendment) Bill, 2020.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1705381>

Q.80) Ans: B

Exp:

- **India-Uzbekistan conducts joint military exercise DUSTLIK II.** Aim of the Exercise is counter Insurgency (CI) and Counter Terrorism (CT) operations in mountainous, rural and urban scenarios.
- **Vajra Prahar** is the joint military exercise between the special forces of **India and the United States (US)**. The aim of Vajra Prahar is to enhance military cooperation between the two countries and also to capitalise on the rich repository of experiences of each other's armies.
- **Nomadic elephant is a joint military exercise between India-Mongolia** (not Kazakhstan). Aimed at training troops in counter insurgency & counter terrorism operations under UN mandat.
- **Sampriti is a joint military exercise between India-Bangladesh.** Aimed to enhance military cooperation between the two countries.
- Recently in news because, In march India conducted the second edition of joint military exercise "Dustlik-2" started in Chaubatia, Uttarakhand.

Source:

<https://pib.gov.in/PressReleasePage.aspx?PRID=1703755>

Q.81) Ans: A

Exp:

Quasars are very luminous objects in faraway galaxies that emit jets at radio frequencies.

About the Discovery- It is named P172+18, the quasar emitted wavelengths which had a redshift of 6.8.

Gravitational redshift occurs as particles of light (photons) climb out of a gravitational well like a black hole and the light's wavelength gets drawn out. This shifts the wavelength to the red part of the light spectrum - hence "redshift". It took 13 billion years for the quasar's light to reach earth.

NEWS : Recently, an international team of astronomers has discovered the most distant 'Radio-Loud' Quasar with the help of European Southern Observatory's Very Large Telescope (ESO's VLT).

SOURCE : Onlyias

Q.82) Ans: D

Exp:

- **Statement 1 is correct:** In a Westminster parliamentary system, the defeat of an appropriation bill in a parliamentary vote generally necessitates either the resignation of a government or the calling of a general election. One of the more famous examples of the defeat of a supply bill was the 1975 Australian constitutional crisis, when the Senate, which was controlled by the opposition, refused to approve a package of appropriation and loan bills, prompting Governor-General Sir John Kerr to dismiss Prime Minister Gough Whitlam and appoint Malcolm Fraser as caretaker Prime Minister until the next election (where the Fraser government was elected).
- **Statement 2 is correct:** As per article 114 of the Constitution, the government can withdraw money from the Consolidated Fund only after receiving approval from Parliament.
- **Statement 3 is correct:** The government introduces the Appropriation Bill in the lower house of Parliament after discussions on Budget proposals and Voting on Demand for Grants. The Appropriation Bill is first passed by the Lok Sabha and then sent to the Rajya Sabha. The Rajya Sabha has the power to recommend any amendments in this Bill.

However, it is the prerogative of the Lok Sabha to either accept or reject the recommendations made by the upper house of Parliament.

NEWS : Recently, the Lok Sabha cleared the Appropriation Bill, allowing the Central government to draw funds from the Consolidated Fund of India.

SOURCE : ONLYIAS

Q.83) Ans: B

Exp:

The Various Mudras of The Buddha-

- **Dharmachakra Mudra** - It is also called as the gesture of 'Teaching of the Wheel of Dharma' that describes one of the most important moments in the Buddha's life as he performed the Dharmachakra mudra in his first sermon in Sarnath after he attained enlightenment. It is performed with the help of both the hands which are held against the chest, the left facing inward, covering the right facing outward.
- **Dhyan Mudra** - It is also known as Samadhi or Yoga Mudra. It is performed with the help of two hands, which are placed on the lap and place the right hand on the left hand with stretched fingers (thumbs facing upwards and other fingers of both the hands resting on each other.) This is the characteristic gesture of Buddha Shakyamuni, Dhyan Buddha Amitabh and the Medicine Buddha.
- **Pair 1 is incorrectly matched: Anjali Mudra** -
- **It is also called Namaskara Mudra or Hridayanjali Mudra** that represents the gesture of greeting, prayer and adoration. It is performed by pressing the palms of the hands together in which the hands are held at the heart chakra with thumbs resting lightly against the sternum.
- **Pair 2 is correctly matched: Karana Mudra** - It signifies the warding off of evil which is performed by raising the index and the little finger, and folding the other fingers. It helps in reducing sickness or negative thoughts.
- **Pair 3 is correctly matched: Varada Mudra** - This mudra represents the offering, welcome, charity, giving, compassion and sincerity. It is performed with the help of both the hands in which the palm of right hand is facing forward and fingers extended and left hand palm placed near omphalos with extended fingers.

- **Pair 4 is correctly matched: Vajra Mudra** - This gesture denotes the fiery thunderbolt that symbolises the five elements—air, water, fire, earth, and metal. It is performed with the help of the right fist and left forefinger, which is placed by enclosing the erect forefinger of the left hand in the right fist with the tip of the right forefinger touching (or curled around) the tip of the left forefinger.



NEWS : A 900 year old Monastery was recently unearthed by Archaeological Survey of India (ASI).

SOURCE: <https://www.newindianexpress.com/lifestyle/spirituality/2018/may/27/the-various-mudras-of-the-buddha-1819040.html>

Q.84) Ans: a

Exp:

- The Union Cabinet has approved the Pradhan Mantri Swasthya Suraksha Nidhi (PMSSN).
- **Statement 1 is Correct** : It is a single non-lapsable reserve fund for Health expenditure.
 - It is fund for Health in the Public Account, from the proceeds of Health and Education cess levied under Finance Act, 2007.
 - In the budget speech 2018, existing 3% Education cess is replaced by 4% Health and Education Cess
 - Administration and maintenance of the PMSSN is entrusted to Ministry of Health & Family Welfare (MoHFW).
- **Statement 2 is Incorrect** : In any financial year, the expenditure on schemes of the MoHFW

would be initially incurred from the PMSSN and thereafter, from Gross Budgetary Support (GBS).

- GBS is the government's support for the central plan which is earmarked for meeting the planned outlays of the Central government during the financial year. It includes the tax receipts and other sources of revenue raised by the Government

Q.85) Ans: b

Exp:

- **Statement 1 is Incorrect** : Recently, United Nations General Assembly has declared 2023 as the International Year of Millets. In 2018, India had proposed to celebrate 2023 as International Year of Millets at Food and Agriculture Organization
- **Statement 2 is Correct** : Millets are a group of small-seeded grasses, widely grown as cereal crops or grains for human food and as fodder. They are classified into Major Millets and Minor Millets based on their grain size.
- **Statement 3 is Correct** : Millets are Photo-insensitive (do not require a specific photoperiod for flowering) & resilient to climate change. Millets have low carbon and water footprint (rice plant needs at least 3 times more water to grow in comparison to millets).
- **Statement 4 is Correct** : India is the largest producer of millets in the world with a 41.0% global market share. India produced 11.5 million tonnes of millets in 2020. During 2017-18, the maximum area under millets was in Rajasthan, followed by Maharashtra and Karnataka.

Q.86) Ans: b

Exp:

- Recently, U.S. India Artificial Intelligence (USIAI) has launched for working in the field of Artificial Intelligence (AI)
- **Statement 1 is Correct** : U.S. India Artificial Intelligence (USIAI) is an initiative of IUSSTF.
 - USIAI will serve as a platform to discuss opportunities, challenges, and barriers for bilateral Artificial Intelligence (AI) R&D collaboration, enabling AI innovation, sharing ideas for

developing an AI workforce and recommending modes and mechanisms for catalyzing partnerships.

- It will provide breakthrough solutions in a range of sectors, like agriculture, energy, climate change, affordable housing and smart cities etc.
- **Statement 2 is Incorrect** : INDIAai is the National AI Portal of India - a central hub for everything AI in India and beyond was joint initiative of Ministry of Electronics and IT, National e-Governance Division and NASSCOM.
 - National Association of Software and Services Companies (NASSCOM) has started 'Future Skills Prime initiative' which aims to foster an ecosystem that enhances India's digital talent with the right set of skills through remote and self-paced learning
- **Statement 3 is Incorrect** : The Department of Science & Technology, Governments of India and the U.S. Department of States are respective nodal departments for IUSSTF.

Q.87) Ans: a

Exp:

- **Statement 1 is Correct** : Objective of the Scheme Pradhan Mantri Matru Vandana Yojana are
 - To provide partial compensation for the wage loss in terms of cash incentives so that the woman can take adequate rest before and after delivery of the first living child
 - To improve health seeking behaviour amongst Pregnant Women and Lactating Mothers (PW&LM) through cash incentives
- **Statement 2 is Correct** : All Pregnant Women and Lactating Mothers (PW&LM), excluding
 - PW&LM who are in regular employment with the Central Government or the State Governments or PSUs or
 - those who are in receipt of similar benefits under any law for the time being in force.

A beneficiary is eligible to receive benefits under the scheme only once. In case of miscarriage or still birth, the beneficiary would be eligible to claim the remaining installment(s) in event of any future pregnancy

- **Statement 3 is Incorrect** : Cash incentive of Rs 5000 in three instalments i.e. 1st instalment of Rs 1000/- on early registration of pregnancy, 2nd instalment of Rs 2000/- after six months of pregnancy and 3rd instalment of Rs 2000/- after child birth is registered and the child has received the first cycle of BCG, OPV, DPT and Hepatitis - B, or its equivalent/ substitute.
 - The eligible beneficiaries would receive the incentive given under the Janani Suraksha Yojana (JSY) for Institutional delivery and the incentive received under JSY would be accounted towards maternity benefits so that on an average a woman gets Rs 6000 /

Q.88) Ans: c

Exp:

- **Statement 1 is Correct**: Earth hour 2021 was celebrated on March 27th, 2021 under the theme, "Climate change to Save Earth." This year's event highlighted the link between destruction of nature and increasing outbreaks of diseases like Covid-19. According to data, Delhi saved 334 MW electricity during Earth Hour.
- **Statement 2 is Incorrect** : It is Organized by World Wildlife Fund (WWF), the one-hour lights out event (8:30 - 9:30 pm) is held worldwide toward end of March.
- **Statement 3 is Correct** : It encourages individuals, households, communities and businesses to turn off their non-essential lights for one hour as a symbol for their commitment to the planet.
 - It started as a light out event in Sydney, Australia in 2007 and since then has garnered support in 185 countries.

Q.89) Ans: a

Exp:

- **Statement 1 is Correct** : Aluminium-air batteries utilise oxygen in the air which reacts with an aluminium hydroxide solution to oxidise the aluminium and produce electricity.
 - Metal-air batteries are equipped with a metal anode and an air-breathing cathode through a suitable electrolyte.
 - Apart from aluminium-air, other metal-air batteries that have been extensively studied are sodium (Na)-air, potassium (K)-air, zinc (Zn)-air, magnesium (Mg)-air etc.
- **Statement 2 is Incorrect** : Benefit of Aluminium air battery over Lithium-ion battery are
 - Significantly cheaper, lighter and more energy-dense alternative.
 - Expected to offer much greater range of 400 km or more per battery compared to lithium-ion batteries (150-200 kilometres) per full charge.
 - Unlike other batteries, aluminium plate in an aluminium-air battery can be recycled and traded directly for industrial uses.
 - **But**, One of the key downsides of aluminium-air batteries is that they cannot be recharged like lithium-ion batteries
- **Statement 2 incorrect**: The Hon'ble High Court of Delhi, in **Babloo Chauhan @ Dabloo v. State Government of NCT of Delhi**, while acknowledging the **inadequacy of the legislative framework for providing relief to the victims of wrongful prosecution and incarceration, requested the Law Commission of India to "undertake a comprehensive examination of the issue.**
 - Pursuant thereto, after extensive examination, the Law Commission submitted its Report titled as, **"Wrongful Prosecution (Miscarriage of Justice): Legal Remedy"** to the Government of India for its consideration.
 - Besides providing for the standards to be applied in cases of "miscarriage of justice" and the meaning of "wrongful prosecution¹⁴", certain amendment¹⁵ to the Code of Criminal Procedure, 1973 ("CrPC/Code") have also been suggested in the said Report.

Extra edge by OnlyIas

- **Hussainara Khatoon (I) v. Home Secretary State of Bihar**: The Hon'ble Supreme Court in Hussainara Khatoon (I) v. Home Secretary State of Bihar, (1980) 1 SCC 81, recognizing the **right to speedy trial as one of the offshoots of the rights conferred under Article 21 of the Constitution.**
 - Holding that, **"No procedure which does not ensure a reasonably quick trial can be regarded as "reasonable, fair or just" and it would fall foul of Article 21."**

Source:

<https://www.thehindu.com/news/national/wrongful-prosecution-plea-in-supreme-court-seeks-framing-implementation-of-guidelines/article34044500.ece>
<https://www.mondaq.com/india/trials-appeals-compensation/777794/wrongful-prosecution-victim39s-rights-and-state39s-obligations>
<https://www.prsindia.org/content/wrongful-prosecution-miscarriage-justice-legal-remedies>

Q.90) Ans: D

Exp:

- **Statement 1 incorrect**: Article 21 of the Constitution of India confers, on every person, the **fundamental rights of life and personal liberty.**
 - As per this Article, **"No person shall be deprived of his life or personal liberty except according to procedure established by law."**
 - The Hon'ble Supreme Court in **Maneka Gandhi v. Union of India, (1978)** while re-defining the contour of Article 21 of Constitution had, **inter alia**, held that the **procedure envisaged under this Article must be "right, just and fair" and not "arbitrary, fanciful or oppressive".**

Q.91) Ans: A

Exp:

- **Statement 1 correct:** The OCI Card has proved to be very popular amongst foreigners of Indian Origin and spouses of foreign origin of Indian citizens or OCI cardholders, as it helps them in hassle free entry and unlimited stay in India.

- So far about 37.72 lakh OCI Cards have been issued by the Government of India.
- As per the extant law, a foreigner of Indian origin or a foreign spouse of an Indian citizen or foreign spouse of an Overseas Citizen of India (OCI) cardholder, can be registered as an OCI cardholder.
- OCI card is a life-long visa for entry into and stay in India with a number of other major benefits attached to it which are not available to other foreigners.

- **Statement 2 incorrect:** With a view to facilitate the OCI cardholders, it has now been decided by the Government of India to dispense with this requirement.

- A person who has got registration as OCI cardholder prior to attaining the age of 20 years will have to get the OCI card re-issued only once when a new passport is issued after his/her completing 20 years of age, so as to capture his/ her facial features on attaining adulthood.
- If a person has obtained registration as an OCI cardholder after attaining the age of 20 years, there will be no requirement for the re-issue of the OCI card.

- **Statement 3 incorrect:** In response to persistent demands for "dual citizenship" particularly from the Diaspora in North America and other developed countries and keeping in view the Government's deep commitment towards fulfilling the aspirations and expectations of Overseas Indians,
 - The Overseas Citizenship of India (OCI) Scheme was introduced by amending the Citizenship Act, 1955 in

August 2005. The Scheme was launched during the Pravasi Bharatiya Divas convention 2006 at Hyderabad.

- The Overseas Citizens of India card is issued to people of Indian origin globally which gives them almost all the privileges of an Indian national except for the right to vote, government service and buying agricultural land.

Source:

<https://pib.gov.in/PressReleaselframePage.aspx?PRID=1712082>

<https://www.thehindu.com/news/national/oci-card-holders-no-longer-required-to-carry-old-passports-for-india-travel/article34195384.ece>

Q.92) Ans: B

Exp:

- **Statement 1 incorrect:** The Ministry of Home Affairs' (MHA) cyber crime volunteers programme targets to rope in around 500 persons to flag unlawful content on the Internet for "improvement in the cybercrime ecosystem of India".

- The programme will include 200 "cyber awareness promoters" and 50 "cyber experts", according to the outcome monitoring framework of the MHA's budget 2021-22.
- The output outcome framework developed by the NITI Aayog was adopted in the 2019-20 budget to define and evaluate government schemes below ₹500 crore in terms of outcomes.

- **Statement 2 correct:** The Internet Freedom Foundation (IFF), a digital liberties organisation, has written to the MHA that the scheme will lead to a culture of surveillance and constant suspicion in society creating potential social distrust.

- There is fear in civil society activities that the programme may be misused by certain elements to extract misguided personal or political vendettas.



- It should also be noted that there is no process in place for withdrawal of complaints once submitted.
- **Statement 3 correct:** The National Cybercrime Ecosystem Management Unit is part of the Indian Cyber Crime Coordination Centre (I4C) scheme.
 - It was launched by Union Home Minister Amit Shah in January 2020.
 - There is also a rolling out of the National Cyber Research and Innovation scheme, for institutions to solve the problems of law enforcement agencies.

Source:

<https://www.thehindu.com/news/national/cyber-crime-volunteers-plan-fraught-with-dangers-internet-freedom-foundation/article33973393.ece>
https://www.mha.gov.in/division_of_mha/cyber-and-information-security-cis-division/Details-about-Indian-Cybercrime-Coordination-Centre-I4C-Scheme
<https://www.thehindu.com/news/national/no-centralised-list-under-cybercrime-volunteer-programme-home-ministry/article34409024.ece>

Q.93) Ans: D

Exp:

Option d correct:

- Recently the Prime Minister of India paid tribute to former PM Morarji Desai on his 124th birth anniversary, recalling his principle-based politics. Morarji Desai was born on February 29, 1896 in Bhadeli village, now in the Bulsar district of Gujarat. His father was a school teacher and a strict disciplinarian. From his childhood, young Morarji learnt from his father the value of hard work and truthfulness under all circumstances. He was educated at St. Bursar High School and passed his matriculation examination. After graduating from the Wilson Civil Service of the then Bombay Province in 1918, he served as a Deputy Collector for twelve years.
- In 1930, when India was in the midst of the freedom struggle launched by Mahatma

Gandhi, Shri Desai, having lost his confidence in the British sense of justice, decided to resign from Government service and to plunge into the struggle. Desai was imprisoned thrice during the freedom struggle. He became a Member of the All India Congress Committee in 1931 and was Secretary of the Gujarat Pradesh Congress Committee until 1937.

- Desai was arrested and detained on June 26, 1975, when Emergency was declared. He was kept in solitary confinement and was released on January 18, 1977. He campaigned vigorously throughout the length and breadth of the country and was largely instrumental in achieving the resounding victory of the Janata Party in the General Elections held in March, 1977 for the Sixth Lok Sabha. Shri Desai was selected to the Lok Sabha from the Surat Constituency in Gujarat. He was later unanimously elected as Leader of the Janata Party in Parliament and was sworn in as the Prime Minister of India on March 24, 1977.

Source:

<https://www.thehindu.com/news/national/pm-modi-pays-tribute-to-morarji-desai-on-his-birth-anniversary/article30948950.ece>
https://www.pmindia.gov.in/en/former_pm/shri-morarji-desai-2/

Q.94) Ans: A

Exp:

- **Statement 1 Correct:** Salt production and distribution in India had long been a lucrative monopoly of the British.
 - Through a series of laws, the Indian populace was prohibited from producing or selling salt independently, and instead Indians were required to buy expensive, heavily taxed salt that often was imported.
 - This affected the great majority of Indians, who were poor and could not afford to buy it. Indian protests against the salt tax began in the 19th century and remained a major

contentious issue throughout the period of British rule of the subcontinent.

- **Statement 2 incorrect:** On the morning of April 6, Gandhi and his followers picked up handfuls of salt along the shore, thus technically “producing” salt and breaking the law.
 - No arrests were made that day, and Gandhi continued his satyagraha against the salt tax for the next two months, exhorting other Indians to break the salt laws by committing acts of civil disobedience.
 - Thousands were arrested and imprisoned, including Jawaharlal Nehru in April and Gandhi himself in early May after he informed Lord Irwin (the viceroy of India) of his intention to march on the nearby Dharasana salt works.
- **Statement 3 incorrect:** At Dharasana salt works the march on the saltworks went ahead as planned on May 21, led by the poet Sarojini Naidu, and many of the some 2,500 peaceful marchers were attacked and beaten by police. By the end of the year, some 60,000 people were in jail. Dharasana salt works

Source:

<https://indianexpress.com/article/explained/explained-a-2021-dandi-march-to-mark-75-years-of-independence-7221855/>

Q.95) Ans: C

Exp:

- Recently Avian influenza (H5N1) led to the death of over 5,000 migratory birds in the Pong Dam Wildlife Sanctuary area in Kangra district. As many as 27 migratory birds have been reported dead on account of avian influenza, setting off alarm bells and forcing authorities to shut down the sanctuary.

- **Pong Dam Lake Wildlife Sanctuary:** it is situated in the district of Kangra of Himachal Pradesh with area of Area: 240 sq. km.
 - In 1975, Pong dam was built across the Beas River. It is also called the Pong reservoir or the Maharana Pratap Sagar.
 - The lake is fed by Beas River and its numerous perennial tributaries such as Gaj, Neogal, Binwa, Uhl, Bangana, and Baner.
 - The placid lake harbours around 22 species of fish, including rare fish like sal and gad. The adequate water level of the lake makes it an ideal destination to indulge in water sports.
 - Pong Dam Sanctuary is also a national wetland. In 1975 and It was declared a wildlife sanctuary in 1983, and is now also a Ramsar site.
 - The sanctuary plays host to around 220 species of birds belonging to 54 families.
 - Migratory birds from all over Hindukush Himalayas and also as far as Siberia come here during winter.
 - The sanctuary area is covered with tropical and subtropical forests, which shelters a great number of Indian Wildlife animals.

Source: <https://www.thehindu.com/sci-tech/energy-and-environment/27-migratory-birds-found-dead-at-wildlife-sanctuary/article34184767.ece>

Q.96) Ans: D

Exp:

- **Statements 1 correct:** Bode’s law of planetary distances, proposed in 1766 by the German astronomer Johann D. Titius and popularized by his compatriot Johann E. Bode, who used the scheme to advance the notion of a “missing” planet between Mars and Jupiter for asteroids.
 - The discovery of the planet Uranus in 1781 by the British astronomer William Herschel at a distance that closely fit the distance predicted by Bode’s law was taken as strong evidence of its correctness.

- **Statement 2 correct:** Asteroid, also called minor planet or planetoid, any of a host of small bodies, about 1,000 km (600 miles) or less in diameter, that orbit the Sun primarily between the orbits of Mars and Jupiter in a nearly flat ring called the asteroid belt.

- It is because of their small size and large numbers relative to the major planets that asteroids are also called minor planets.

- **Statement 3 correct:** All asteroids orbit the Sun in elliptical orbits and move in the same direction as the major planets. Some elliptical orbits are very nearly circles, whereas others are highly elongated (eccentric).

- An orbit is completely described by six geometric parameters called its elements. Orbital elements, and hence the shape and orientation of the orbit, also vary with time because each object is gravitationally acting on, and being acted upon by, all other bodies in the solar system.

- **Statement 2 incorrect:** Heat wave a weather phenomena characterized by **low humidity**, and high temperature. which may **exacerbate drought** which may exacerbate the health effects of heat-related stress, which include heat exhaustion, dehydration, and heatstroke.

Extra Edge by Onlyias

Definition of Heat wave: There is no formal, standardized definition of a heat wave.

- The World Meteorological Organization defines it as **five or more consecutive days** during which the daily maximum temperature surpasses the average maximum temperature by 5 °C (9 °F) or more.
- Some countries have adopted their own standards. For example, the India Meteorological Department requires that temperatures increase 5–6 °C (9–10.8 °F) or more above the normal temperature.

Source: <https://www.thehindu.com/sci-tech/science/earth-safe-from-asteroid-for-100-years-nasa/article34182248.ece>

Source: <https://www.thehindu.com/sci-tech/energy-and-environment/arctic-warming-is-causing-heat-waves-in-india-study/article34157092.ece>

Q.97) Ans: D

Exp:

- **Statement 1 incorrect:** Heat wave is a period of prolonged abnormally high surface temperatures relative to those normally expected.

- Heat waves may span several days to several weeks and are significant causes of weather-related mortality, affecting developed and developing countries alike.
- Globally, the increasing frequency and intensity of heat waves observed since the 1950s has been associated with climate change.
- Earth's internal heat : inside the Earth from radioactive decay of elements and residual heat leftover from Earth's formation.

Q.98) Ans: C

Exp:

- **Statement 1 is correct:** Aadhaar is a unique document issued by the Unique Identification Authority of India on behalf of the Government of India. According to the Supreme court's judgement of 2018 on validity of the Aadhaar scheme held that the Aadhaar Act has struck a fair balance between the right of privacy of the individual along with the right to life of the same individual as a beneficiary of government benefits.
- **Statement 2 is correct:** SC upheld the Aadhar for government services by using the Doctrine of Proportionality.
- **Statement 3 is incorrect:** Apex Court upheld the Constitutional validity of the Aadhaar Act but with certain caveats. It allowed the

mandatory linking of Aadhaar for filing tax returns and accessing welfare schemes but **removed the requirement** for bank accounts and SIM cards.

- **Statement 4 is correct:** Aadhaar card is not proof of citizenship, but citizens must provide it to receive welfare payments and social services. It has so far empowered more than 125.79 crore residents with unique digital identity
- **Why in news:** The Supreme Court remarked while hearing a plea which claimed that the technological system based on iris identification, thumbprints, non-functioning of the internet in rural and remote areas, etc., is leading to large scale cancellation of ration cards without notice to the family concerned.

Source: <https://www.hindustantimes.com/india-news/cancellation-of-ration-card-not-linked-to-aadhaar-a-serious-issue-sc-tells-govt-101615975448932.html>

- **Statement 1 is correct:** Himalayan serow resembles a cross between a goat, a donkey, a cow. They are herbivores and are found in the eastern, central, and western Himalayas but not in trans Himalayas. Typically found at altitudes between 2000 meters and 4000 meters.
- **Statement 2 is incorrect:** Himalayan serow was previously assessed as near threatened and now has been vulnerable in the IUCN red list. Recently the first-time Serow was sighted in Assam.
- **Statement 3 is incorrect:** it is listed under schedule I of the wildlife protection act 1972
- **Recently in the news,** because Himalayan serows have experienced a significant decline in population size, range size, and habitat last night this is expected to continue due to intensive human impact (IUCN).



Source: <https://www.thehindu.com/news/national/other-states/in-a-first-himalayan-serow-spotted-in-assam/article33972852.ece#:~:text=A%20couple%20of%20nature%20guides,Tiger%20Reserve%20on%20December%203.&text=%E2%80%99CWe%20have%20now%20confirmed%20the,in%20Manas's%20Bansbari%20Mathanguri%20forest.>

Q.99) Ans: B

Exp:

- **Brute force acquisition** can be performed by 3rd party passcode brute force tools that send a series of passcodes/ passwords to the mobile device. This technique uses trial and error in an attempt to create the correct combination of password or PIN to authenticate access to the mobile device.
- **Mobile device forensic cloning** forms a part of mobile device forensics and is a bit-for-bit copy of an entire mobile device. Some investigating agencies and forensic science laboratories carry out "imaging" or forensic cloning of a mobile phone or any digital device if they believe it would aid the investigation or help prove a case against someone in a court of law.

Source:

<https://indianexpress.com/article/explained/can-data-deleted-from-your-phone-be-recovered-using-forensic-cloning-how-to-prevent-personal-data-misuse-6607853/>

Q.100) Ans: B

Exp: